CAS SciFinder

Task History

Initiating Search

November 10, 2025, 10:07 PM

Search:

CAS Lexicon Search:

Mannich reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (7,602)	**References**	View Results
Exported: Retrieved Related Reaction Results + Filters (1,021) Filtered By: Yield: **90-100%, 80-89%, 70-79%** Document Type: **Journal** Publication Year: **2020 to 2025**	**Reactions**	View Results

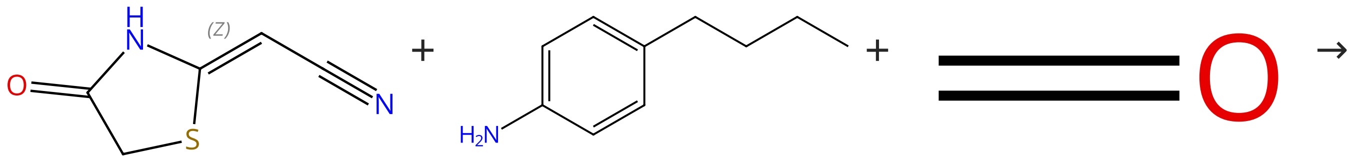
CAS SciFinder

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction)

Steps: **1** Yield: **100%**

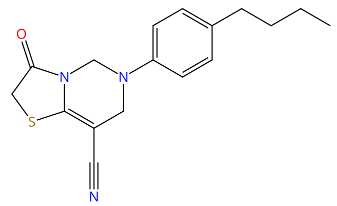


Double bond geometry shown

Suppliers (7)

Suppliers (75)

Suppliers (188)



31-614-CAS-46186149

Steps: **1** Yield: **100%**

1.1 **Solvents:** Ethanol, Water; rt; 4 min, 80 °C

Experimental Protocols

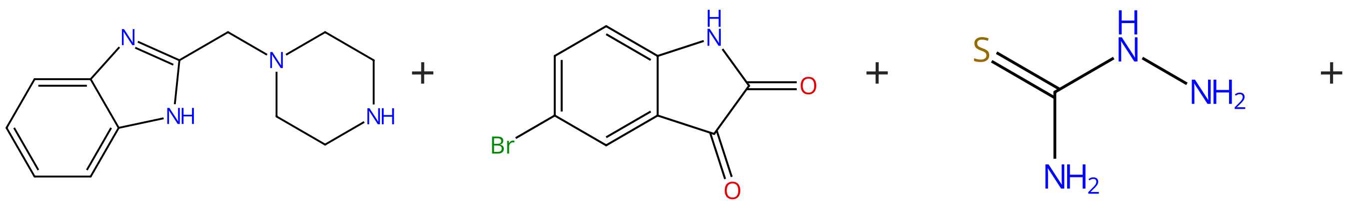
Green Synthesis and Unexpected Transformation of New Oxothiazolo[3,2-c]Pyrimidine Carbonitriles

By: Ozbil, Cagri; et al

Journal of Heterocyclic Chemistry (2025), 62(7), 458-467.

Scheme 2 (1 Reaction)

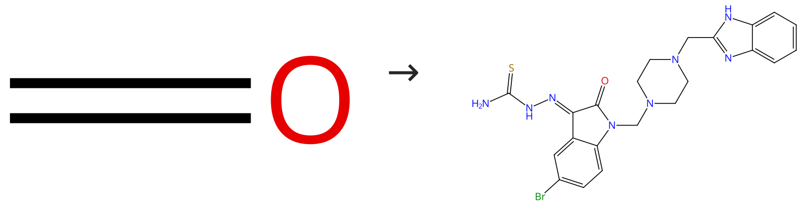
Steps: **1** Yield: **99%**



Suppliers (33)

Suppliers (98)

Suppliers (62)



Suppliers (188)

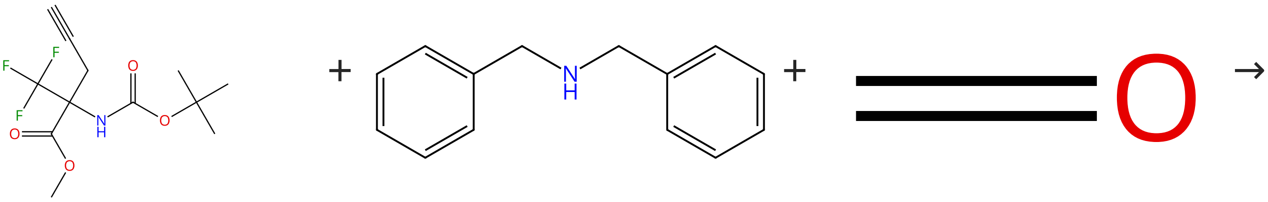
31-310-CAS-22967955	Steps: **1** Yield: **99%**
1.1	**Catalysts:** 4-[[[[[(Dimethylamino)(sulfoimino)methyl] sulfoamino](sulfoimino)methyl]imino]methyl]benzoic acid (Fe3O4 support used) **Solvents:** Ethanolamine; 2 h, reflux
Experimental Protocols	

One-pot multicomponent synthesis of novel 2-(piperazin-1-yl) quinoxaline and benzimidazole derivatives, using a novel sulfamic acid functionalized Fe_3O_4 MNPs as highly effective nanocatalyst

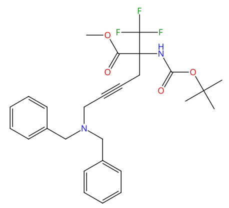
By: Esam, Zohreh; et al

Applied Organometallic Chemistry (2021), 35(1), e6005.

Scheme 3 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (9) Suppliers (91) Suppliers (188)



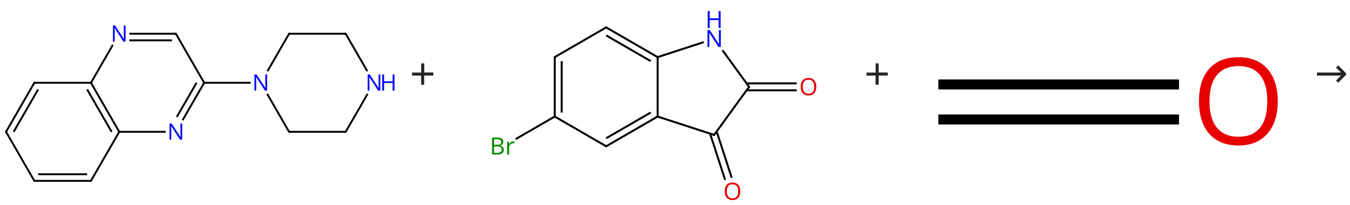
31-614-CAS-32761234	Steps: **1** Yield: **99%**
1.1	**Catalysts:** Cuprous iodide **Solvents:** 1,4-Dioxane; 16 h, 90 °C
Experimental Protocols	

Synthesis of α-CF_3-substituted γ, δ-didehydro lysine deriva tives

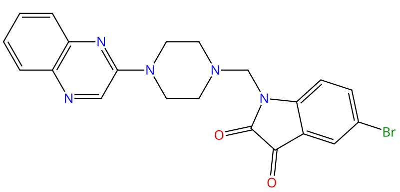
By: Philippova, Anna N.; et al

Mendeleev Communications (2022), 32(2), 260-261.

Scheme 4 (2 Reactions) Steps: **1** Yield: **98-99%**



Suppliers (31) Suppliers (98) Suppliers (188)



31-315-CAS-22975799	Steps: **1** Yield: **99%**
1.1	**Catalysts:** 4-[[[[[(Dimethylamino)(sulfoimino)methyl] sulfoamino](sulfoimino)methyl]imino]methyl]benzoic acid (Fe3O4 support used) **Solvents:** Ethanolamine; 3 h, reflux
Experimental Protocols	

One-pot multicomponent synthesis of novel 2-(piperazin-1-yl) quinoxaline and benzimidazole derivatives, using a novel sulfamic acid functionalized Fe_3O_4 MNPs as highly effective nanocatalyst

By: Esam, Zohreh; et al

Applied Organometallic Chemistry (2021), 35(1), e6005.

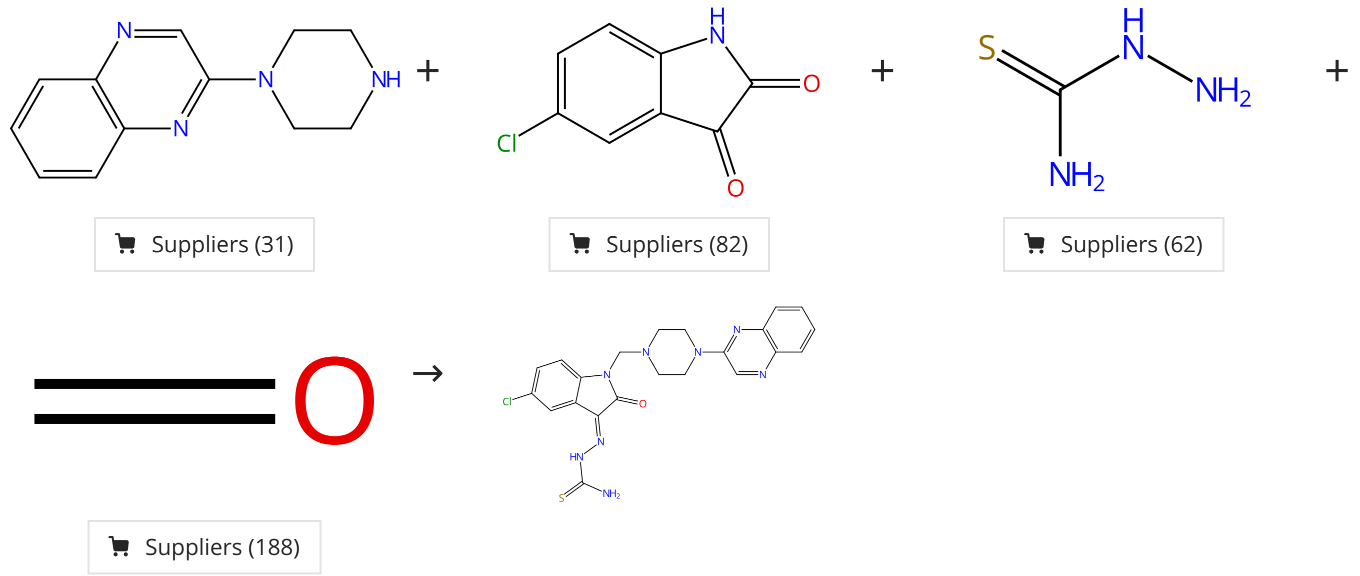
31-614-CAS-37559175	Steps: **1** Yield: **98%**
1.1	**Catalysts:** Benzenesulfonamide, 4-amino-*N*-[imino[[3-(trimethoxysilyl)propyl]amino]methyl]- (sulfonated and aluminum iron oxide supported) **Solvents:** Ethanol; 3 h, reflux

Green synthesis, anti-proliferative evaluation, docking, and MD simulations studies of novel 2-piperazinyl quinoxaline derivatives using hercynite sulfaguanidine-SA as a highly efficient and reusable nanocatalyst

By: Esam, Zohreh; et al

RSC Advances (2023), 13(36), 25229-25245.

Scheme 5 (2 Reactions) Steps: **1** Yield: **99%**



31-614-CAS-37559196	Steps: **1** Yield: **99%**
1.1	**Catalysts:** Benzenesulfonamide, 4-amino-*N*-[imino[[3-(trimethoxysilyl)propyl]amino]methyl]- (sulfonated and aluminum iron oxide supported) **Solvents:** Ethanol; 3 h, reflux

Green synthesis, anti-proliferative evaluation, docking, and MD simulations studies of novel 2-piperazinyl quinoxaline derivatives using hercynite sulfaguanidine-SA as a highly efficient and reusable nanocatalyst

By: Esam, Zohreh; et al

RSC Advances (2023), 13(36), 25229-25245.

31-310-CAS-22969362	Steps: **1** Yield: **99%**
1.1	**Catalysts:** 4-[[[[[(Dimethylamino)(sulfoimino)methyl]sulfoamino](sulfoimino)methyl]imino]methyl]benzoic acid (Fe3O4 support used) **Solvents:** Ethanolamine; 2 h, reflux
Experimental Protocols	

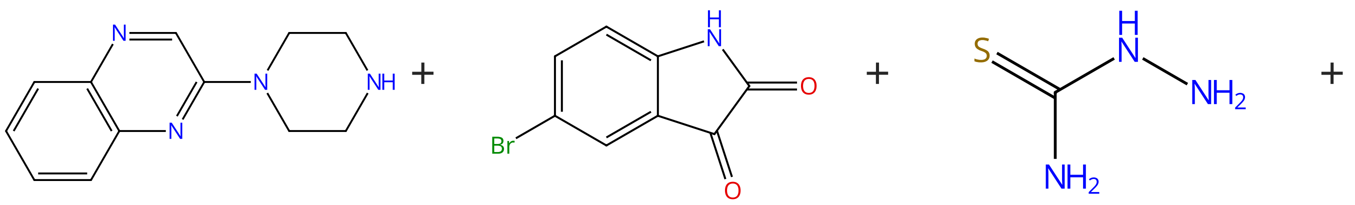
One-pot multicomponent synthesis of novel 2-(piperazin-1-yl) quinoxaline and benzimidazole derivatives, using a novel sulfamic acid functionalized Fe_3O_4 MNPs as highly effective nanocatalyst

By: Esam, Zohreh; et al

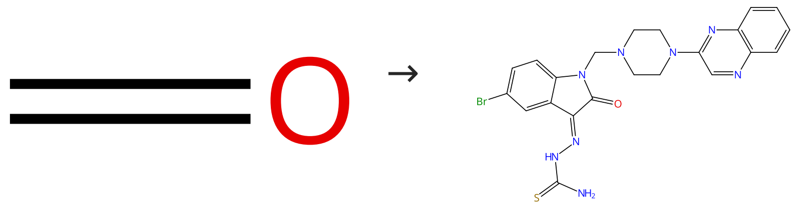
Applied Organometallic Chemistry (2021), 35(1), e6005.

Scheme 6 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (31) Suppliers (98) Suppliers (62)



Suppliers (188)

31-310-CAS-22969036

Steps: **1** Yield: **99%**

1.1 **Catalysts:** 4-[[[[[(Dimethylamino)(sulfoimino)methyl]sulfoamino](sulfoimino)methyl]imino]methyl]benzoic acid (Fe3O4 support used)
Solvents: Ethanolamine; 2 h, reflux

Experimental Protocols

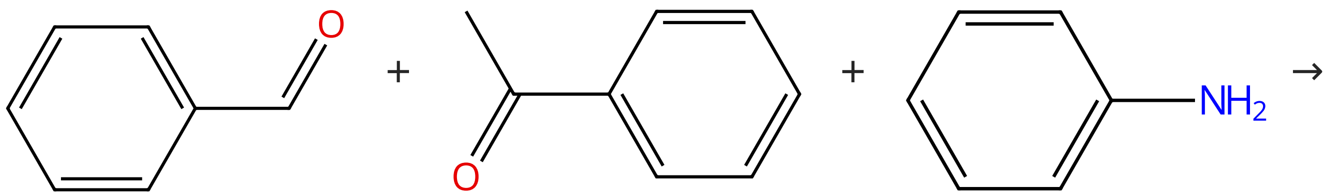
One-pot multicomponent synthesis of novel 2-(piperazin-1-yl) quinoxaline and benzimidazole derivatives, using a novel sulfamic acid functionalized Fe_3O_4 MNPs as highly effective nanocatalyst

By: Esam, Zohreh; et al

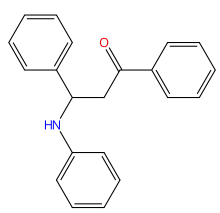
Applied Organometallic Chemistry (2021), 35(1), e6005.

Scheme 7 (2 Reactions)

Steps: **1** Yield: **97-99%**



Suppliers (69) Suppliers (105) Suppliers (113)



Suppliers (7)

31-315-CAS-22822313

Steps: **1** Yield: **99%**

1.1 **Catalysts:** Tungstate(3-), tetracosa-μ-oxododecaoxo[μ12-[phosphato(3-)-κO:κO:κO:κO':κO':κO':κO'':κO'':κO'':κO''':κO''':κO''']]dodeca-, hydrogen, hydrate (1:3:?) (SBA-15); 1.8 h, rt

Experimental Protocols

$H_3PW_{12}O_{40}$ anchored on three-dimensional and networked S BA-15 as efficient and recyclable catalyst for Mannich reaction

By: Zhang, Tengfei; et al

Journal of the Mexican Chemical Society (2020), 64(1), 1-13.

31-614-CAS-32700037	Steps: **1** Yield: **97%**
1.1	1 h, rt
1.2	**Reagents:** Water

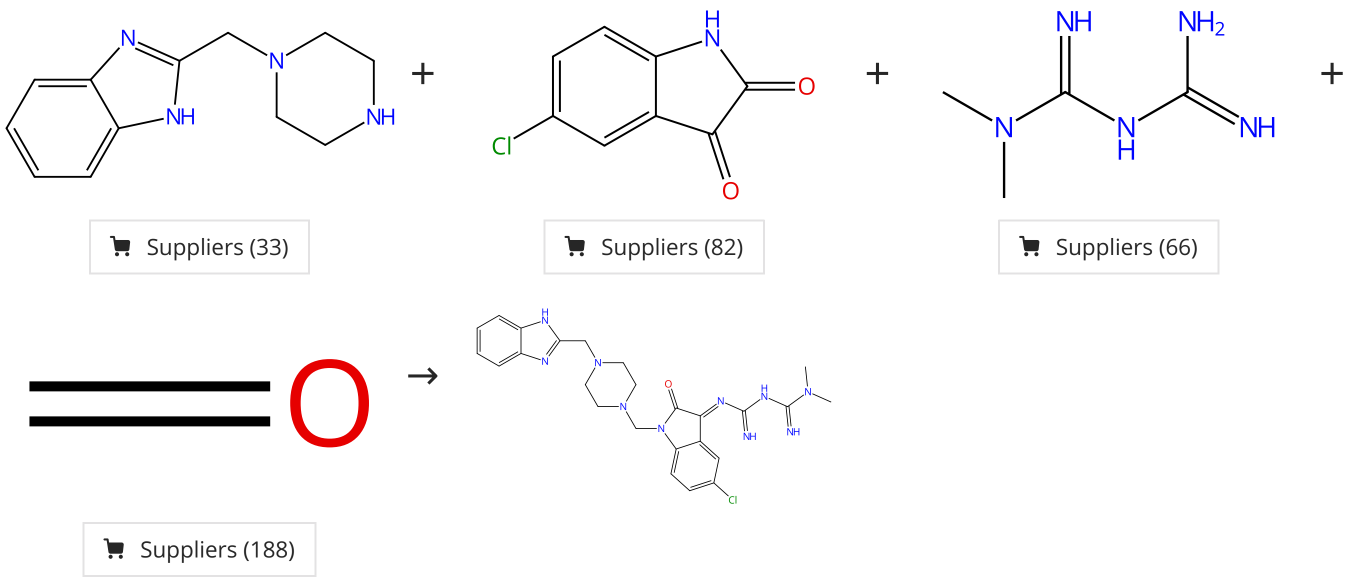
Experimental Protocols

Aluminized Polyborate: A New and Eco-friendly Catalyst for One-pot Multicomponent Synthesis of 1,3-Diaryl-3-(arylamino)propan-1-ones via Mannich Reaction

By: Mali, Anil S.; et al

Organic Preparations and Procedures International (2022), 54(4), 338-345.

Scheme 8 (1 Reaction) Steps: **1** Yield: **98%**



31-309-CAS-22969799	Steps: **1** Yield: **98%**
1.1	**Catalysts:** 4-[[[[[(Dimethylamino)(sulfoimino)methyl] sulfoamino](sulfoimino)methyl]imino]methyl]benzoic acid (Fe3O4 support used) **Solvents:** Ethanolamine; 2 h, reflux

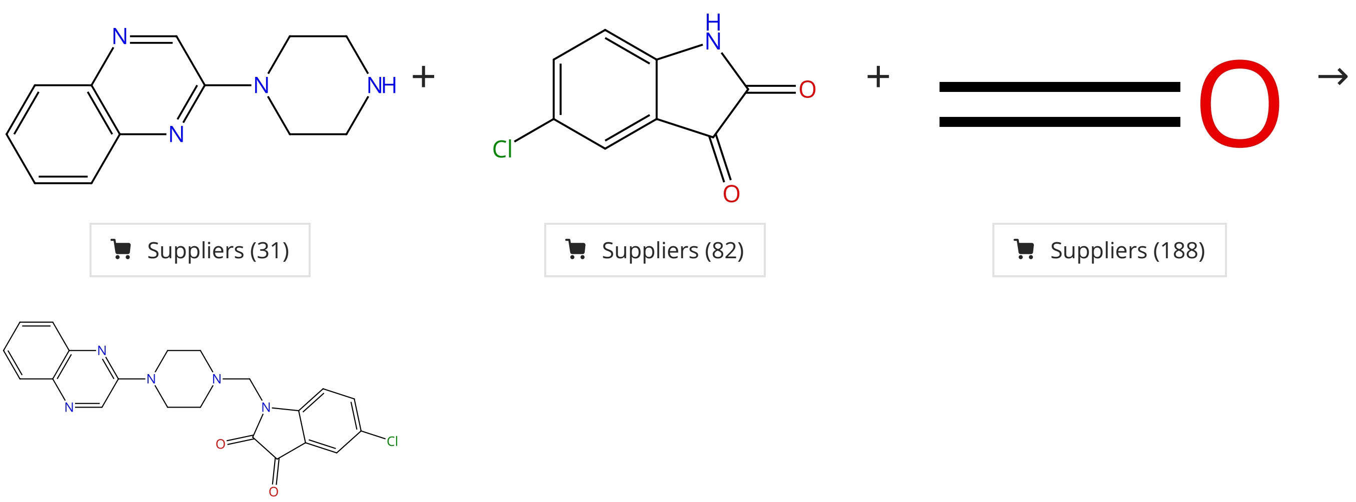
Experimental Protocols

One-pot multicomponent synthesis of novel 2-(piperazin-1-yl) quinoxaline and benzimidazole derivatives, using a novel sulfamic acid functionalized Fe_3O_4 MNPs as highly effective nanocatalyst

By: Esam, Zohreh; et al

Applied Organometallic Chemistry (2021), 35(1), e6005.

Scheme 9 (2 Reactions) Steps: **1** Yield: **98%**



31-614-CAS-37559173	Steps: **1** Yield: **98%**
1.1 **Catalysts:** Benzenesulfonamide, 4-amino-*N*-[imino[[3-(trimethoxysilyl)propyl]amino]methyl]- (sulfonated and aluminum iron oxide supported) **Solvents:** Ethanol; 3 h, reflux	

Green synthesis, anti-proliferative evaluation, docking, and MD simulations studies of novel 2-piperazinyl quinoxaline derivatives using hercynite sulfaguanidine-SA as a highly efficient and reusable nanocatalyst

By: Esam, Zohreh; et al

RSC Advances (2023), 13(36), 25229-25245.

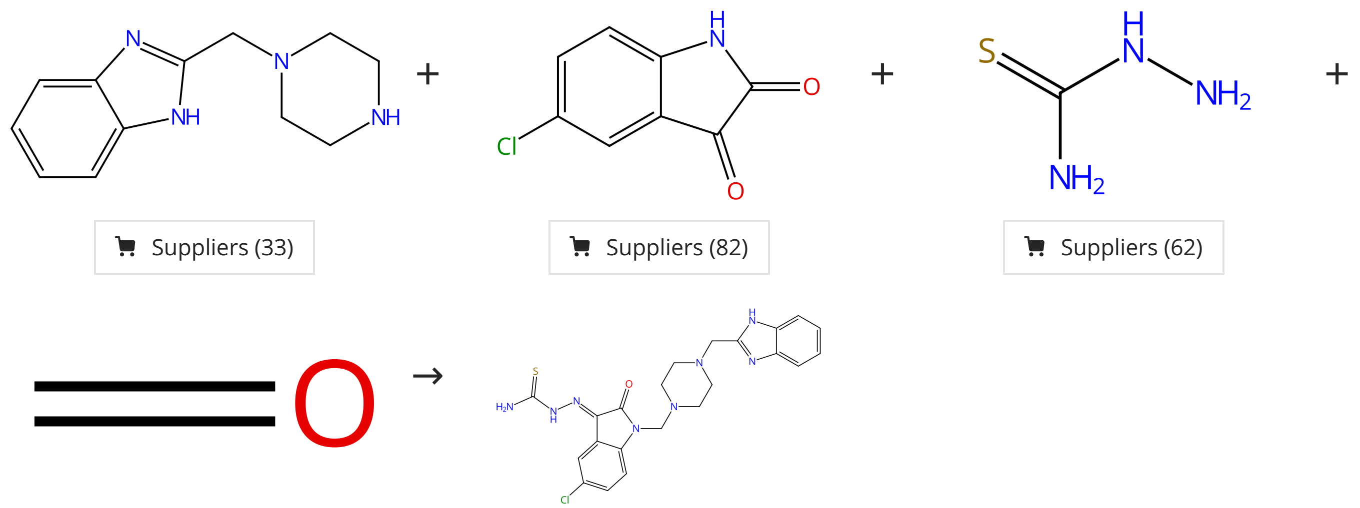
31-315-CAS-22969458	Steps: **1** Yield: **98%**
1.1 **Catalysts:** 4-[[[[[(Dimethylamino)(sulfoimino)methyl]sulfoamino](sulfoimino)methyl]imino]methyl]benzoic acid (Fe3O4 support used) **Solvents:** Ethanolamine; 3 h, reflux	
Experimental Protocols	

One-pot multicomponent synthesis of novel 2-(piperazin-1-yl) quinoxaline and benzimidazole derivatives, using a novel sulfamic acid functionalized Fe_3O_4 MNPs as highly effective nanocatalyst

By: Esam, Zohreh; et al

Applied Organometallic Chemistry (2021), 35(1), e6005.

Scheme 10 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (33) Suppliers (82) Suppliers (62)

Suppliers (188)

31-310-CAS-22971811	Steps: **1** Yield: **98%**
1.1 **Catalysts:** 4-[[[[[(Dimethylamino)(sulfoimino)methyl]sulfoamino](sulfoimino)methyl]imino]methyl]benzoic acid (Fe3O4 support used) **Solvents:** Ethanolamine; 2 h, reflux	
Experimental Protocols	

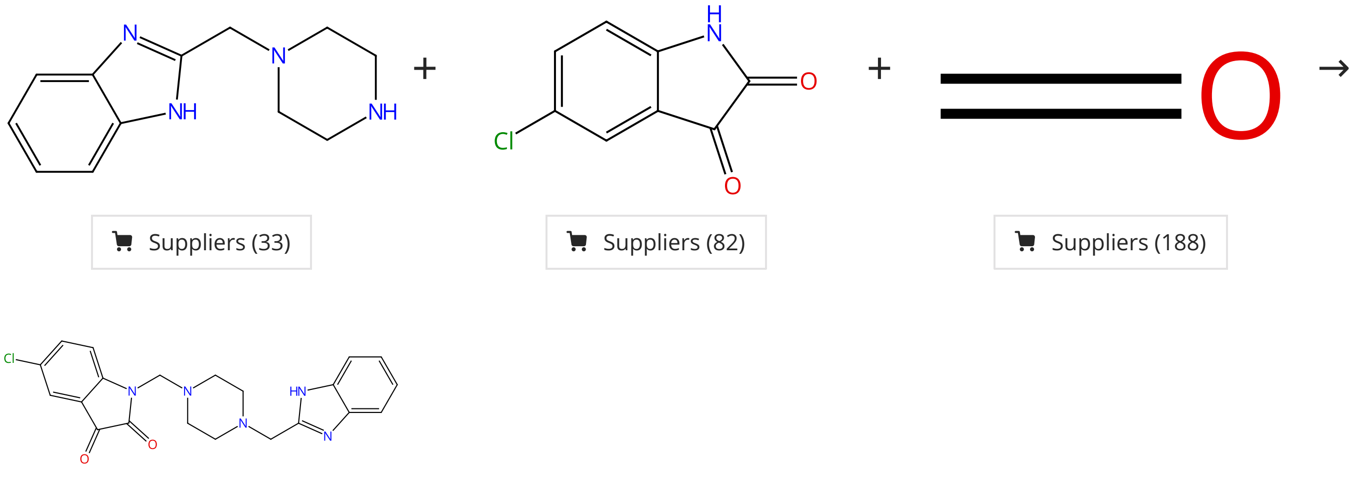
One-pot multicomponent synthesis of novel 2-(piperazin-1-yl) quinoxaline and benzimidazole derivatives, using a novel sulfamic acid functionalized Fe_3O_4 MNPs as highly effective nanocatalyst

By: Esam, Zohreh; et al

Applied Organometallic Chemistry (2021), 35(1), e6005.

Scheme 11 (1 Reaction)

Steps: **1** Yield: **98%**



31-315-CAS-22970937

Steps: **1** Yield: **98%**

1.1 **Catalysts:** 4-[[[[[(Dimethylamino)(sulfoimino)methyl]sulfoamino](sulfoimino)methyl]imino]methyl]benzoic acid (Fe3O4 support used)
Solvents: Ethanolamine; 3 h, reflux

Experimental Protocols

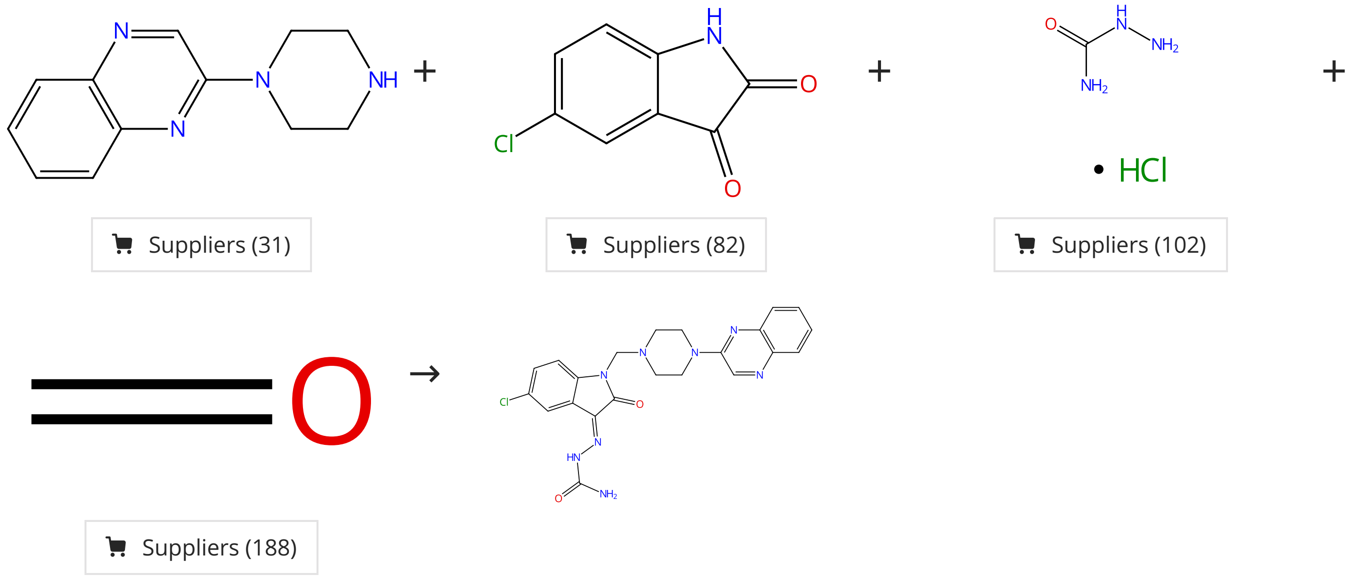
One-pot multicomponent synthesis of novel 2-(piperazin-1-yl) quinoxaline and benzimidazole derivatives, using a novel sulfamic acid functionalized Fe_3O_4 MNPs as highly effective nanocatalyst

By: Esam, Zohreh; et al

Applied Organometallic Chemistry (2021), 35(1), e6005.

Scheme 12 (1 Reaction)

Steps: **1** Yield: **98%**



31-310-CAS-22971748

Steps: **1** Yield: **98%**

1.1 **Catalysts:** 4-[[[[[(Dimethylamino)(sulfoimino)methyl]sulfoamino](sulfoimino)methyl]imino]methyl]benzoic acid (Fe3O4 support used)
Solvents: Ethanolamine; 2 h, reflux

Experimental Protocols

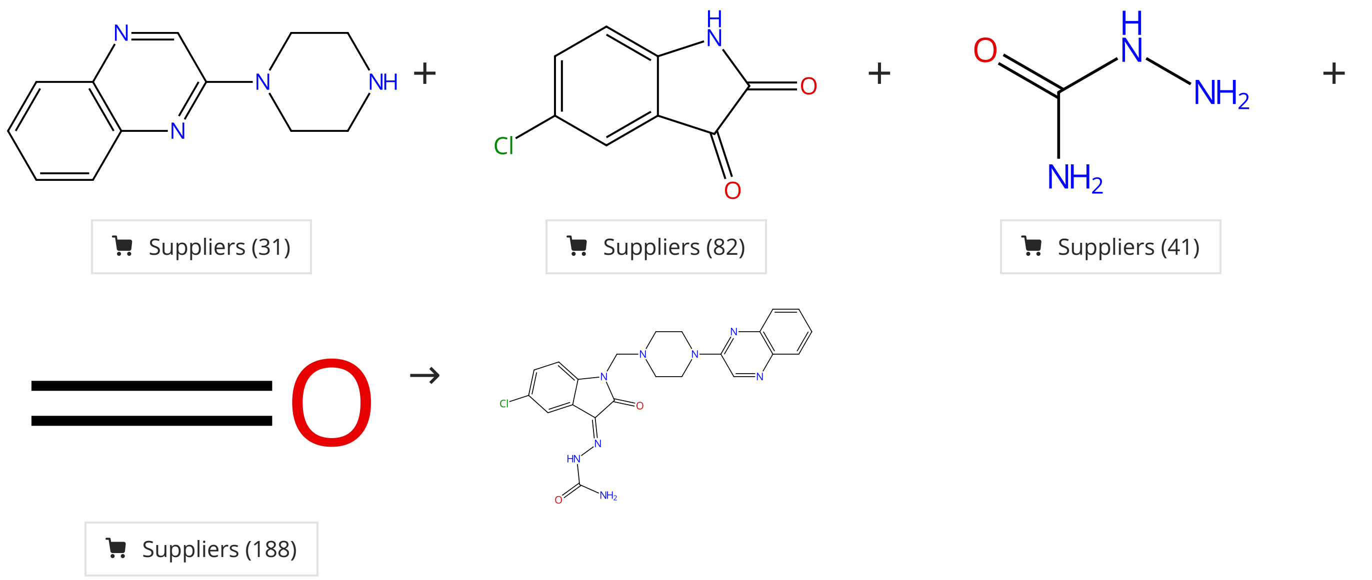
One-pot multicomponent synthesis of novel 2-(piperazin-1-yl) quinoxaline and benzimidazole derivatives, using a novel sulfamic acid functionalized Fe_3O_4 MNPs as highly effective nanocatalyst

By: Esam, Zohreh; et al

Applied Organometallic Chemistry (2021), 35(1), e6005.

Scheme 13 (1 Reaction)

Steps: **1** Yield: **98%**



31-614-CAS-37559179

Steps: **1** Yield: **98%**

1.1 **Catalysts:** Benzenesulfonamide, 4-amino-*N*-[imino[[3-(trimethoxysilyl)propyl]amino]methyl]- (sulfonated and aluminum iron oxide supported)
Solvents: Ethanol; 3 h, reflux

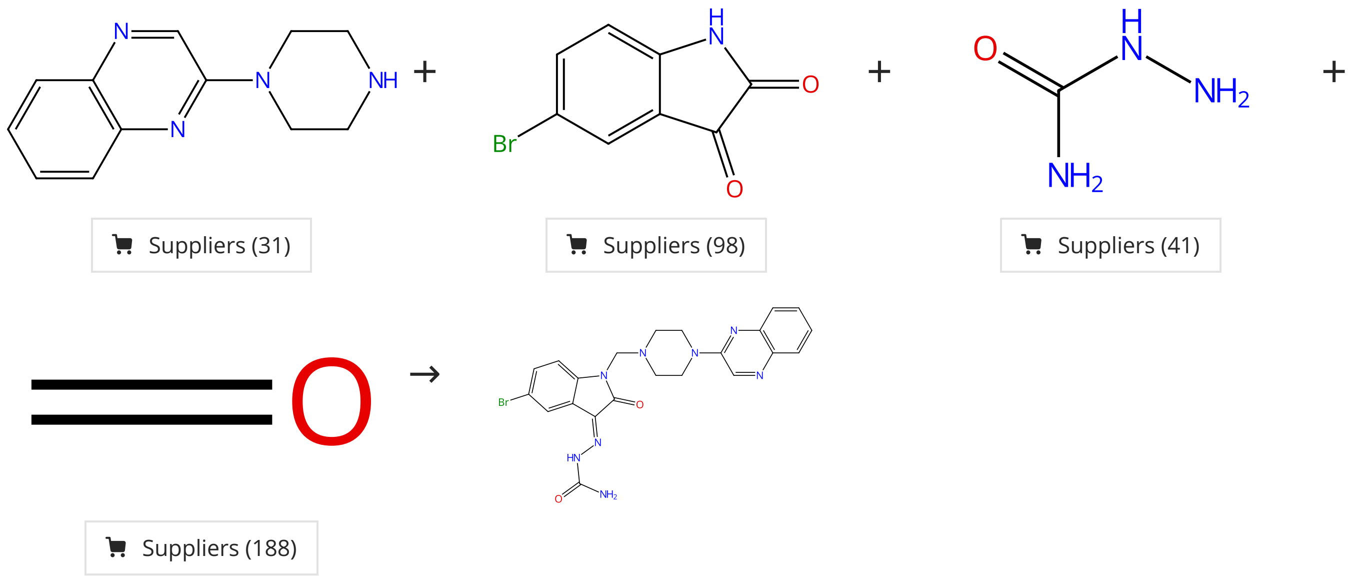
Green synthesis, anti-proliferative evaluation, docking, and MD simulations studies of novel 2-piperazinyl quinoxaline derivatives using hercynite sulfaguanidine-SA as a highly efficient and reusable nanocatalyst

By: Esam, Zohreh; et al

RSC Advances (2023), 13(36), 25229-25245.

Scheme 14 (1 Reaction)

Steps: **1** Yield: **98%**



31-614-CAS-37559182

Steps: **1** Yield: **98%**

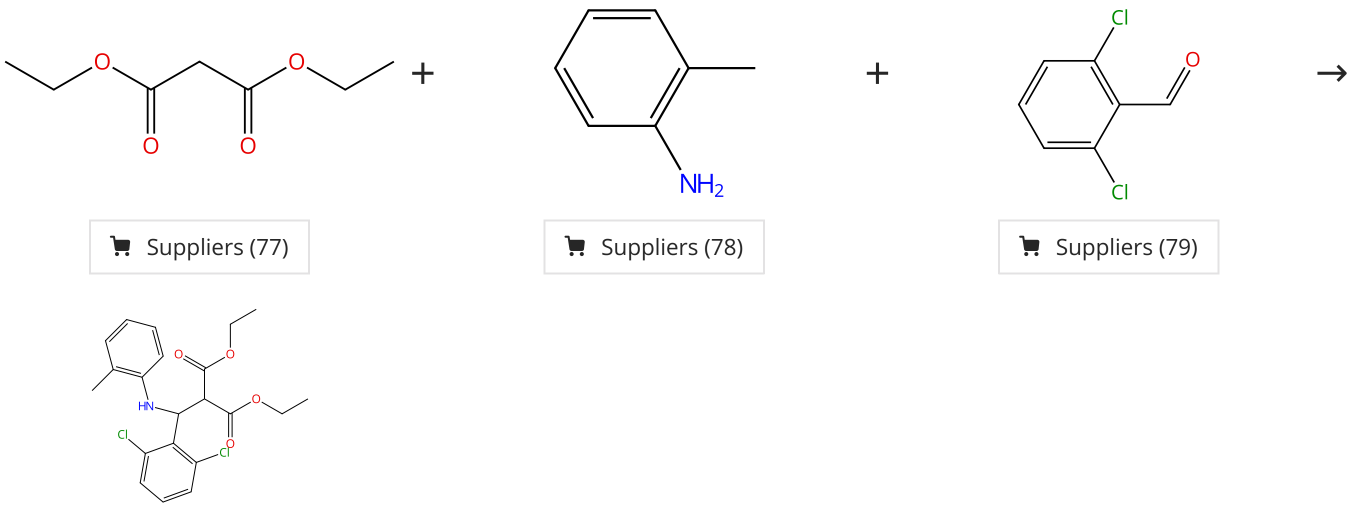
1.1 **Catalysts:** Benzenesulfonamide, 4-amino-*N*-[imino[[3-(trimethoxysilyl)propyl]amino]methyl]- (sulfonated and aluminum iron oxide supported)
Solvents: Ethanol; 3 h, reflux

Green synthesis, anti-proliferative evaluation, docking, and MD simulations studies of novel 2-piperazinyl quinoxaline derivatives using hercynite sulfaguanidine-SA as a highly efficient and reusable nanocatalyst

By: Esam, Zohreh; et al

RSC Advances (2023), 13(36), 25229-25245.

Scheme 15 (1 Reaction) Steps: **1** Yield: **98%**



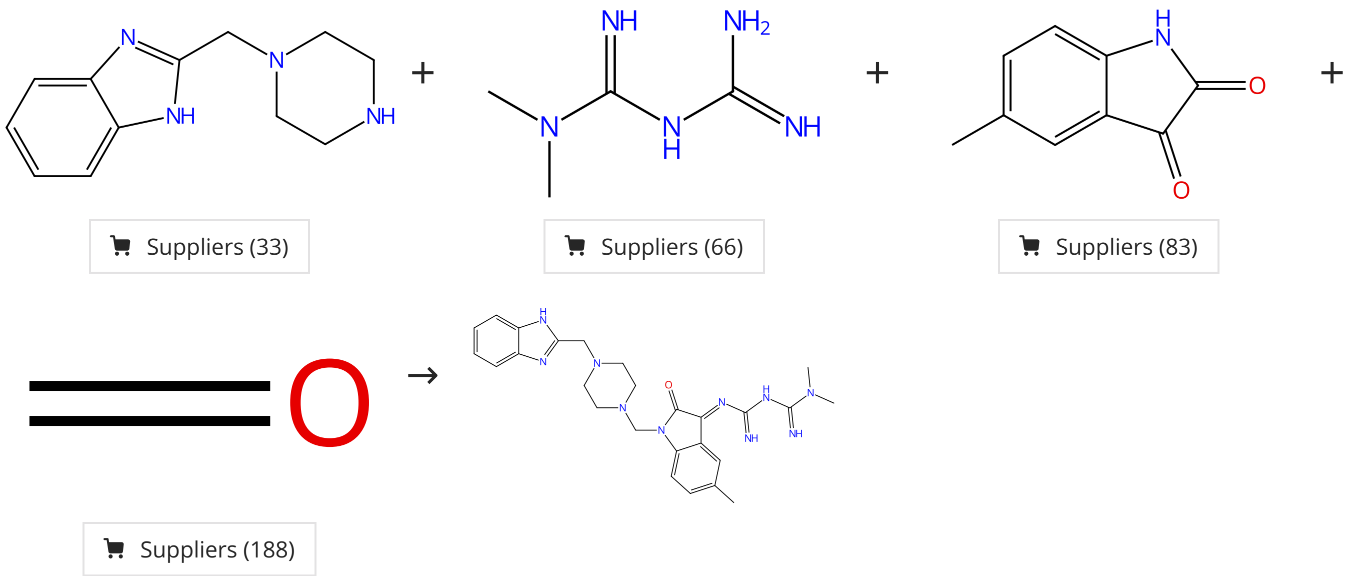
31-315-CAS-23764545	Steps: **1** Yield: **98%**
1.1	**Catalysts:** Ammonium chloride **Solvents:** Ethanol; 10 h, rt
Experimental Protocols	

NH$_4$Cl Catalyzed synthesis of β-amino Esters

By: Waghmare, Smita R.

Results in Chemistry (2020), 2, 100036.

Scheme 16 (1 Reaction) Steps: **1** Yield: **98%**



31-309-CAS-22969321	Steps: **1** Yield: **98%**
1.1	**Catalysts:** 4-[[[[[(Dimethylamino)(sulfoimino)methyl]sulfoamino](sulfoimino)methyl]imino]methyl]benzoic acid (Fe3O4 support used) **Solvents:** Ethanolamine; 2 h, reflux
Experimental Protocols	

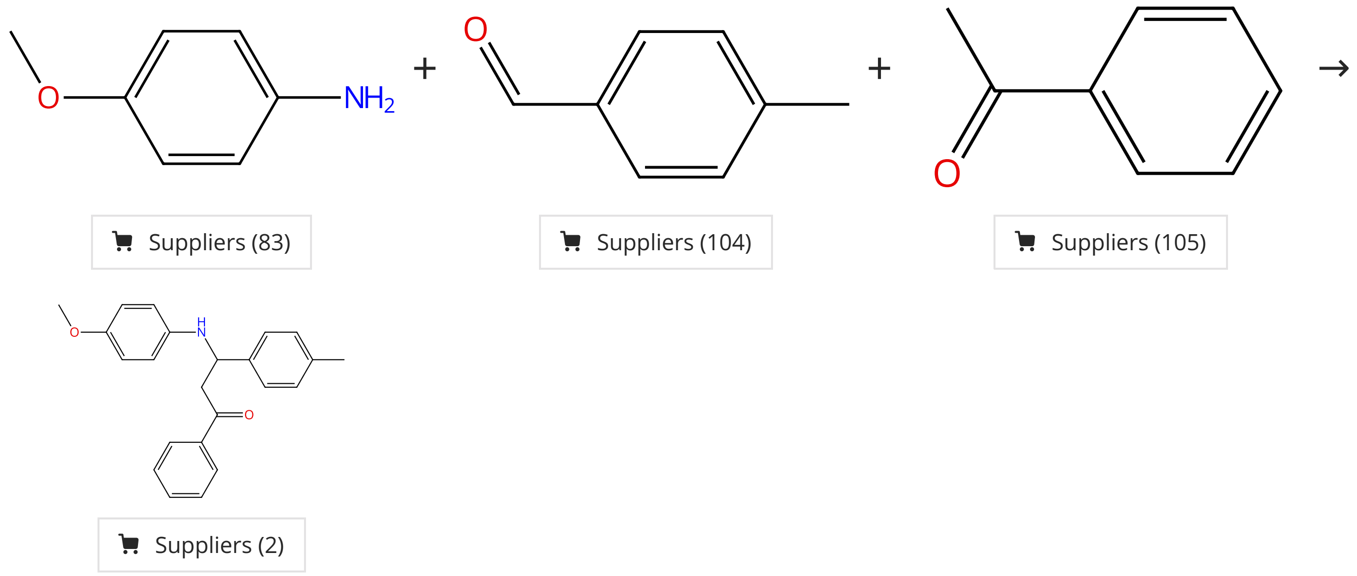
One-pot multicomponent synthesis of novel 2-(piperazin-1-yl) quinoxaline and benzimidazole derivatives, using a novel sulfamic acid functionalized Fe$_3$O$_4$ MNPs as highly effective nanocatalyst

By: Esam, Zohreh; et al

Applied Organometallic Chemistry (2021), 35(1), e6005.

Scheme 17 (1 Reaction)

Steps: **1** Yield: **98%**



31-315-CAS-21619365

Steps: **1** Yield: **98%**

1.1 **Catalysts:** 1*H*-Imidazolium, 1-methyl-3-[2-[[(2*S*)-2-pyrrolidinylcarbonyl]oxy]ethyl]-, 2,2,2-trifluoroacetate (1:1); 2 h, rt

Experimental Protocols

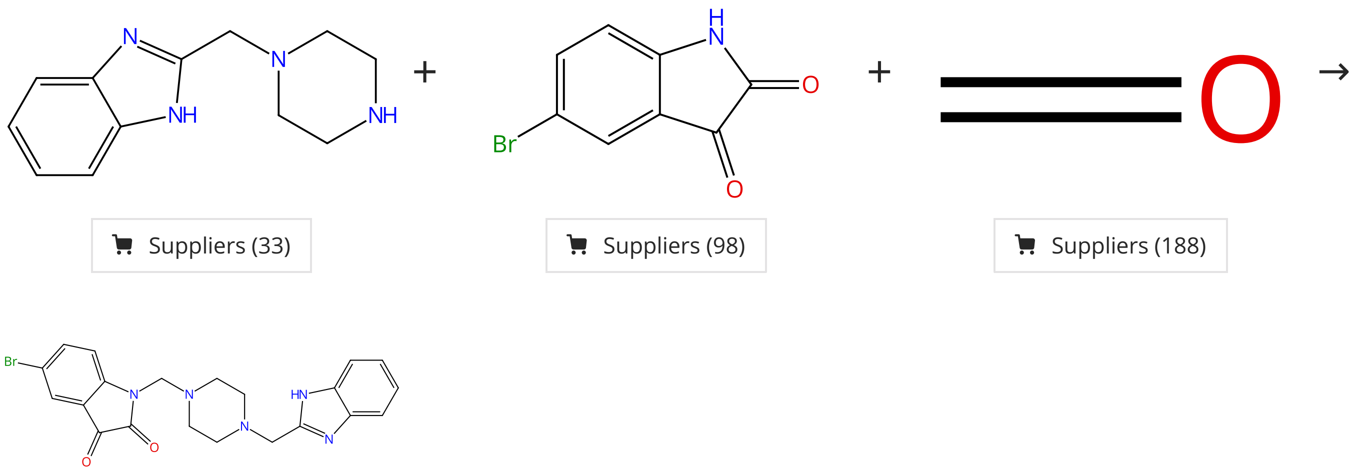
Ionic liquid-immobilized proline(s) organocatalyst-catalyzed one-pot multi-component Mannich reaction under solvent-free condition

By: Prabhakara, M. D.; et al

Research on Chemical Intermediates (2020), 46(4), 2381-2401.

Scheme 18 (1 Reaction)

Steps: **1** Yield: **98%**



31-315-CAS-22969506

Steps: **1** Yield: **98%**

1.1 **Catalysts:** 4-[[[[[(Dimethylamino)(sulfoimino)methyl]sulfoamino](sulfoimino)methyl]imino]methyl]benzoic acid (Fe3O4 support used)
Solvents: Ethanolamine; 3 h, reflux

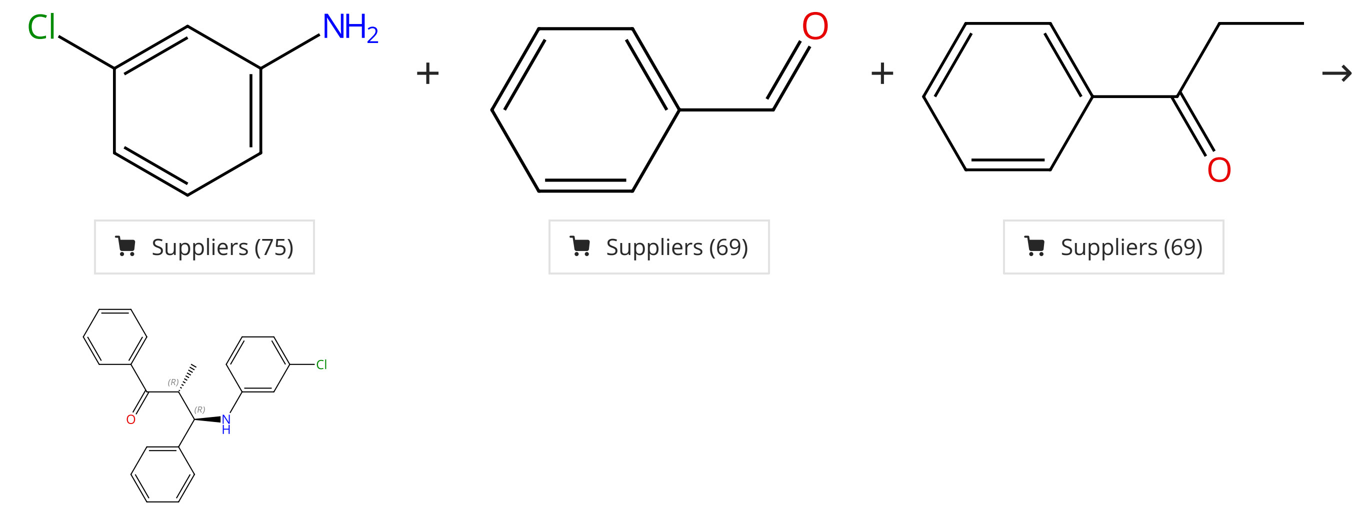
Experimental Protocols

One-pot multicomponent synthesis of novel 2-(piperazin-1-yl) quinoxaline and benzimidazole derivatives, using a novel sulfamic acid functionalized Fe_3O_4 MNPs as highly effective nanocatalyst

By: Esam, Zohreh; et al

Applied Organometallic Chemistry (2021), 35(1), e6005.

Scheme 19 (1 Reaction) Steps: **1** Yield: **98%**



Relative stereochemistry shown

31-614-CAS-24692045 Steps: **1** Yield: **98%**

1.1 **Catalysts:** Ammonium chloride
Solvents: Ethanol; 8 h, 25 °C

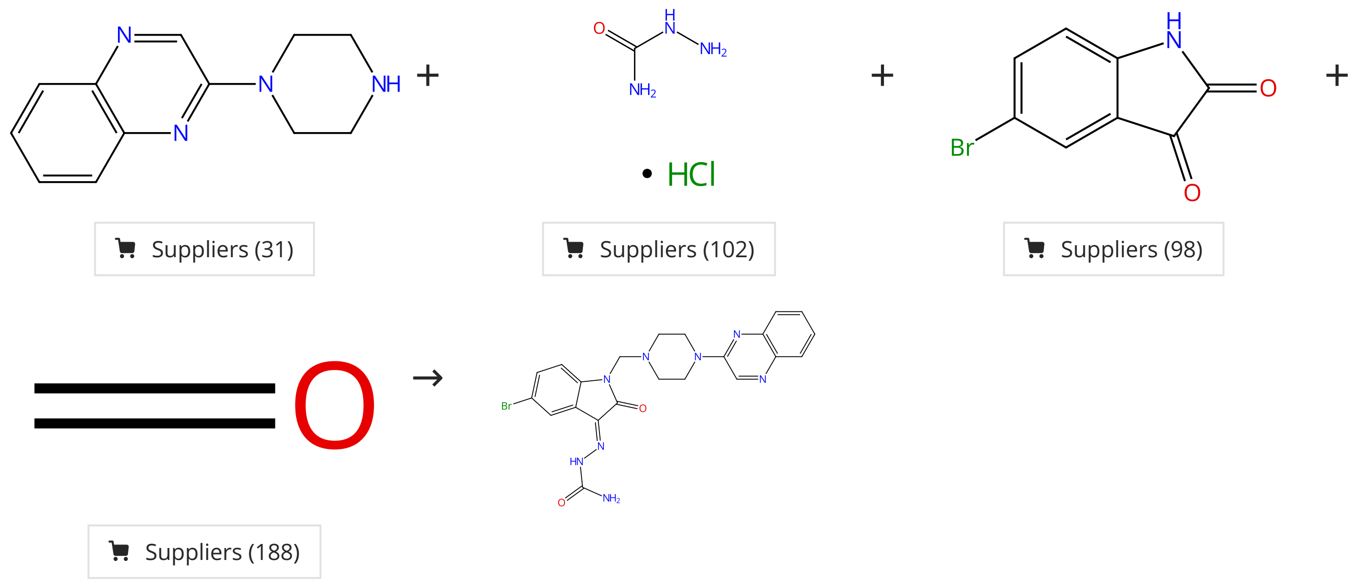
Experimental Protocols

Ammonium Chloride Promoted Synthesis of β-Aminoketones

By: R. Waghmare, Smita; et al

Organic Preparations and Procedures International (2022), 54(1), 65-78.

Scheme 20 (1 Reaction) Steps: **1** Yield: **98%**



31-310-CAS-22971409 Steps: **1** Yield: **98%**

1.1 **Catalysts:** 4-[[[[[(Dimethylamino)(sulfoimino)methyl]sulfoamino](sulfoimino)methyl]imino]methyl]benzoic acid (Fe3O4 support used)
Solvents: Ethanolamine; 2 h, reflux

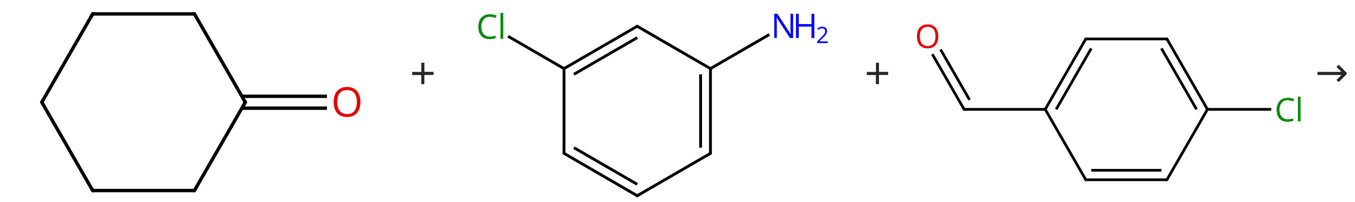
Experimental Protocols

One-pot multicomponent synthesis of novel 2-(piperazin-1-yl) quinoxaline and benzimidazole derivatives, using a novel sulfamic acid functionalized Fe_3O_4 MNPs as highly effective nanocatalyst

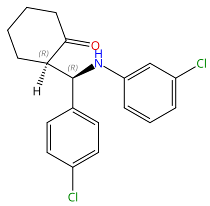
By: Esam, Zohreh; et al

Applied Organometallic Chemistry (2021), 35(1), e6005.

Scheme 21 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (102) Suppliers (75) Suppliers (98)



Relative stereochemistry shown

31-614-CAS-24692046 Steps: **1** Yield: **98%**

1.1 **Catalysts:** Ammonium chloride
Solvents: Ethanol; 8 - 24 h, 25 °C

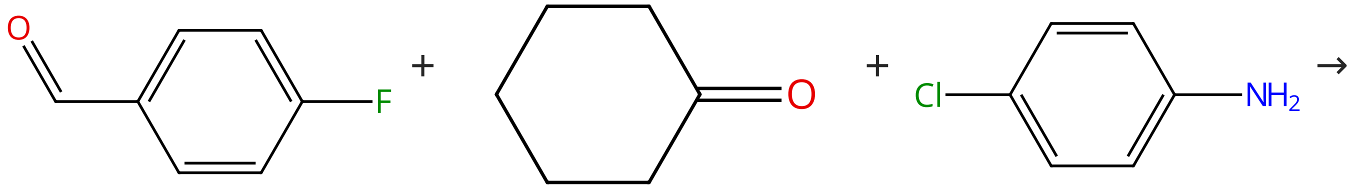
Experimental Protocols

Ammonium Chloride Promoted Synthesis of β-Aminoketones

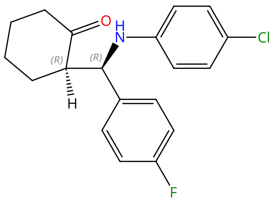
By: R. Waghmare, Smita; et al

Organic Preparations and Procedures International (2022), 54(1), 65-78.

Scheme 22 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (96) Suppliers (102) Suppliers (98)



Relative stereochemistry shown

31-614-CAS-24692030 Steps: **1** Yield: **98%**

1.1 **Catalysts:** Ammonium chloride
Solvents: Ethanol; 8 - 24 h, 25 °C

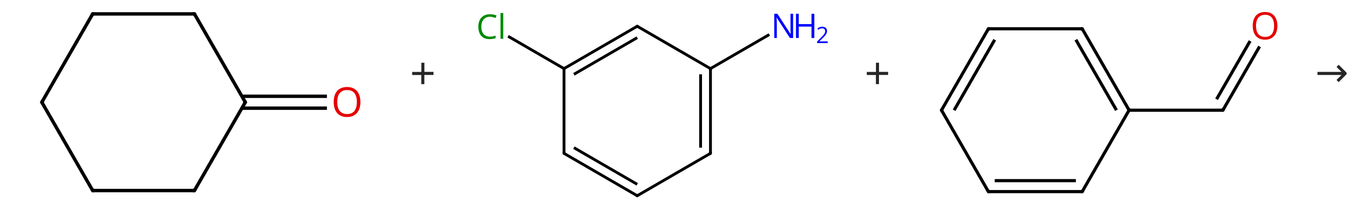
Experimental Protocols

Ammonium Chloride Promoted Synthesis of β-Aminoketones

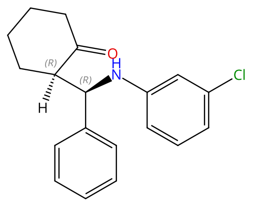
By: R. Waghmare, Smita; et al

Organic Preparations and Procedures International (2022), 54(1), 65-78.

Scheme 23 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (102) Suppliers (75) Suppliers (69)



Relative stereochemistry shown

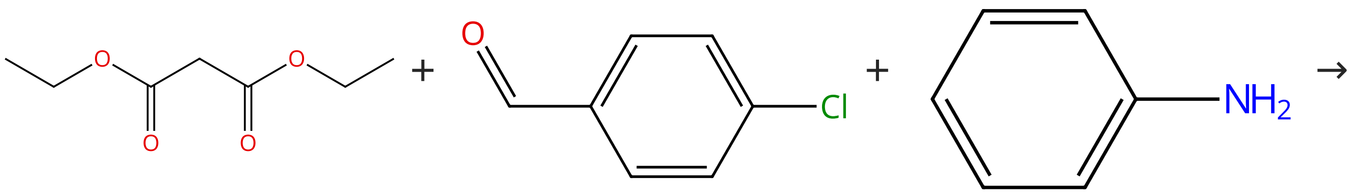
31-614-CAS-24691934	Steps: **1** Yield: **98%**
1.1	**Catalysts:** Ammonium chloride **Solvents:** Ethanol; 8 h, 25 °C
Experimental Protocols	

Ammonium Chloride Promoted Synthesis of β-Aminoketones

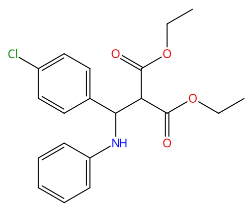
By: R. Waghmare, Smita; et al

Organic Preparations and Procedures International (2022), 54(1), 65-78.

Scheme 24 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (77) Suppliers (98) Suppliers (113)



31-315-CAS-23759748	Steps: **1** Yield: **98%**
1.1	**Catalysts:** Ammonium chloride **Solvents:** Ethanol; 10 h, rt
Experimental Protocols	

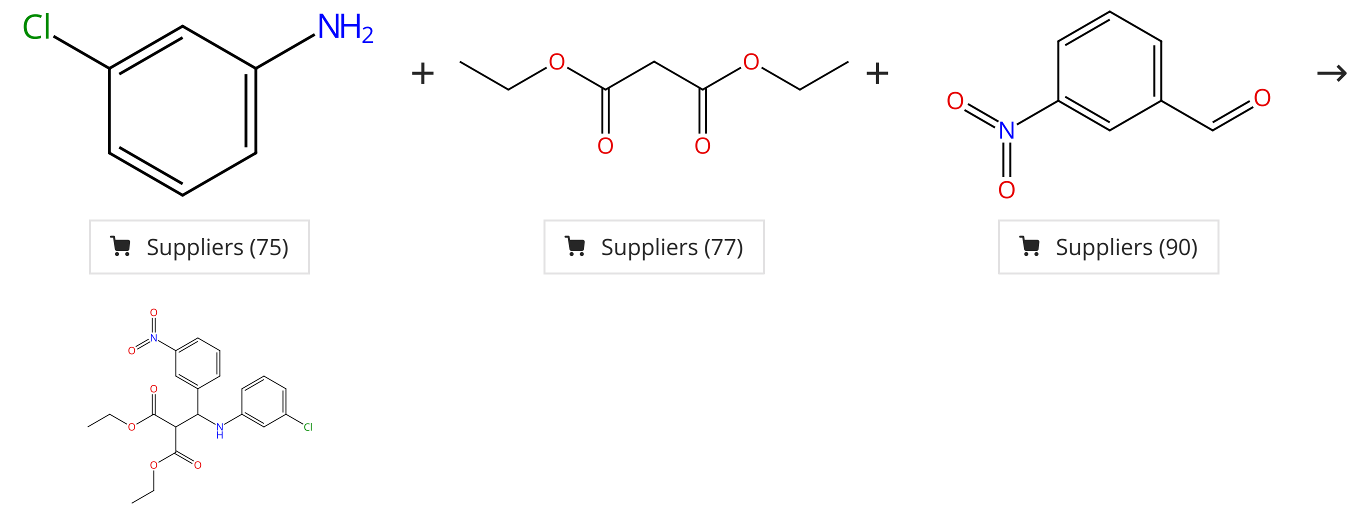
NH_4Cl Catalyzed synthesis of β-amino Esters

By: Waghmare, Smita R.

Results in Chemistry (2020), 2, 100036.

Scheme 25 (1 Reaction)

Steps: **1** Yield: **98%**



31-315-CAS-23757409	Steps: **1** Yield: **98%**
1.1	**Catalysts:** Ammonium chloride **Solvents:** Ethanol; 10 h, rt
Experimental Protocols	

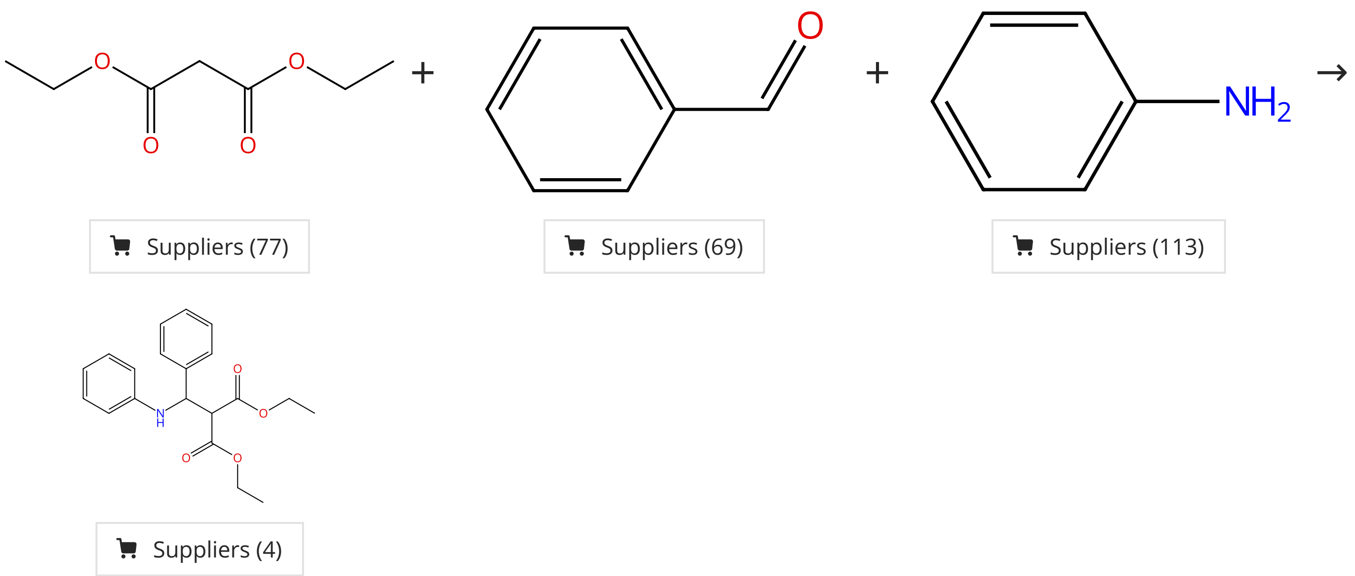
NH_4Cl Catalyzed synthesis of β-amino Esters

By: Waghmare, Smita R.

Results in Chemistry (2020), 2, 100036.

Scheme 26 (1 Reaction)

Steps: **1** Yield: **98%**



31-315-CAS-23760182	Steps: **1** Yield: **98%**
1.1	**Catalysts:** Ammonium chloride **Solvents:** Ethanol; 8 h, rt
Experimental Protocols	

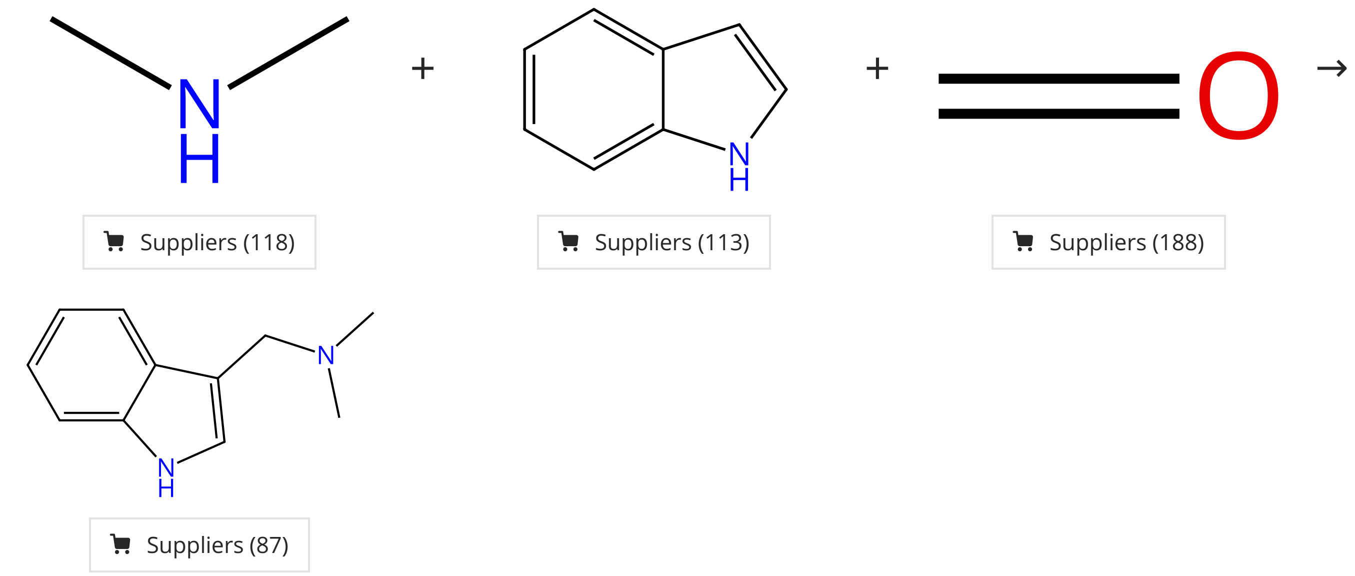
NH_4Cl Catalyzed synthesis of β-amino Esters

By: Waghmare, Smita R.

Results in Chemistry (2020), 2, 100036.

Scheme 27 (1 Reaction)

Steps: **1** Yield: **98%**



31-096-CAS-24404425

Steps: **1** Yield: **98%**

1.1 **Reagents:** Acetic acid
Solvents: Water; 2 h, rt

1.2 **Reagents:** Sodium hydroxide
Solvents: Water; basified, rt

Experimental Protocols

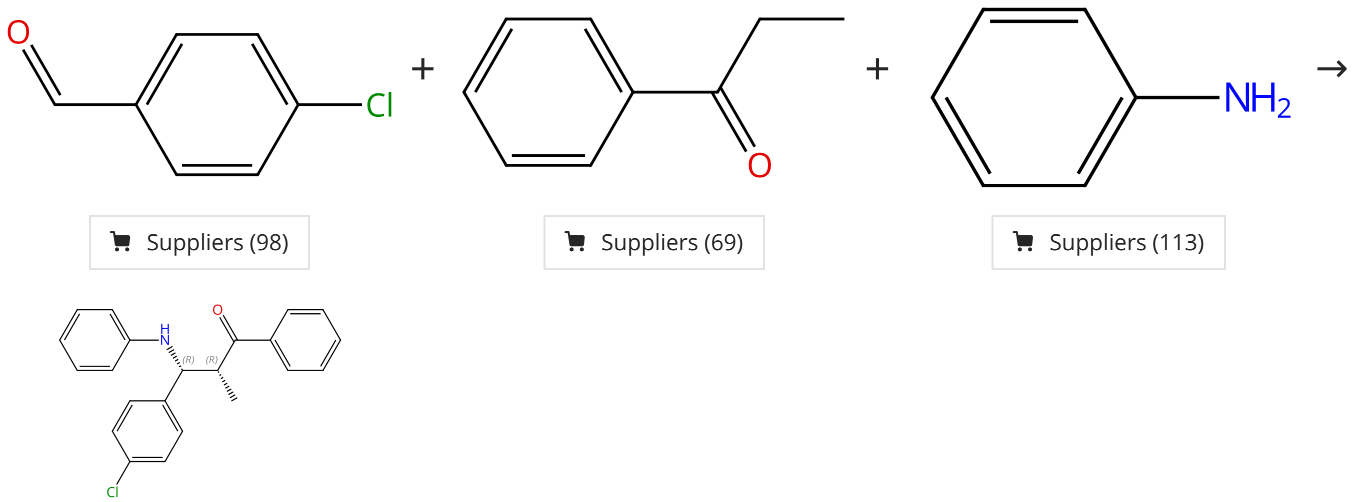
The structural simplification of lysergic acid as a natural lead for synthesizing novel anti-Alzheimer agents

By: Alzweiri, Muhammed; et al

Bioorganic & Medicinal Chemistry Letters (2021), 47, 128205.

Scheme 28 (1 Reaction)

Steps: **1** Yield: **97%**



Relative stereochemistry shown

31-614-CAS-24692104

Steps: **1** Yield: **97%**

1.1 **Catalysts:** Ammonium chloride
Solvents: Ethanol; 8 h, 25 °C

Experimental Protocols

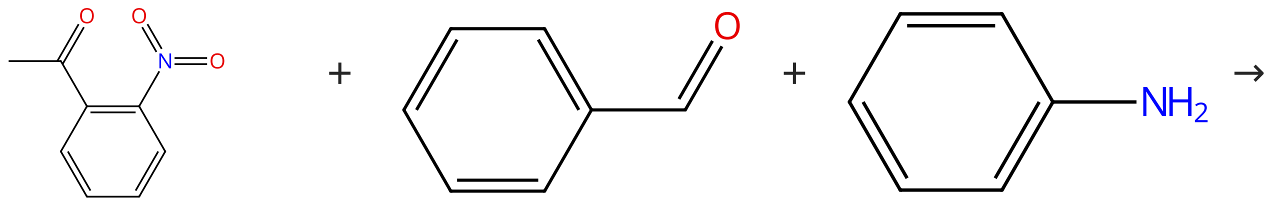
Ammonium Chloride Promoted Synthesis of β-Aminoketones

By: R. Waghmare, Smita; et al

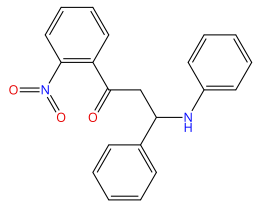
Organic Preparations and Procedures International (2022), 54(1), 65-78.

Scheme 29 (1 Reaction)

Steps: **1** Yield: **97%**



Suppliers (90) | Suppliers (69) | Suppliers (113)



31-315-CAS-21619377

Steps: **1** Yield: **97%**

1.1 **Catalysts:** 1*H*-Imidazolium, 1-methyl-3-[2-[[(2*S*)-2-pyrrolidinylcarbonyl]oxy]ethyl]-, 2,2,2-trifluoroacetate (1:1); 2 h, rt

Experimental Protocols

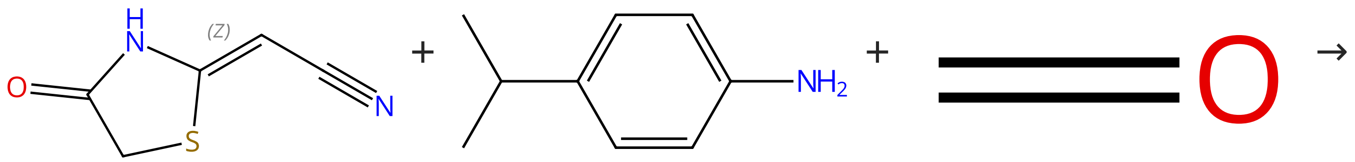
Ionic liquid-immobilized proline(s) organocatalyst-catalyzed one-pot multi-component Mannich reaction under solvent-free condition

By: Prabhakara, M. D.; et al

Research on Chemical Intermediates (2020), 46(4), 2381-2401.

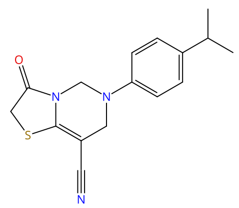
Scheme 30 (1 Reaction)

Steps: **1** Yield: **97%**



Double bond geometry shown

Suppliers (7) | Suppliers (81) | Suppliers (188)



31-614-CAS-46186159

Steps: **1** Yield: **97%**

1.1 **Solvents:** Ethanol, Water; rt; 3 min, 80 °C

Experimental Protocols

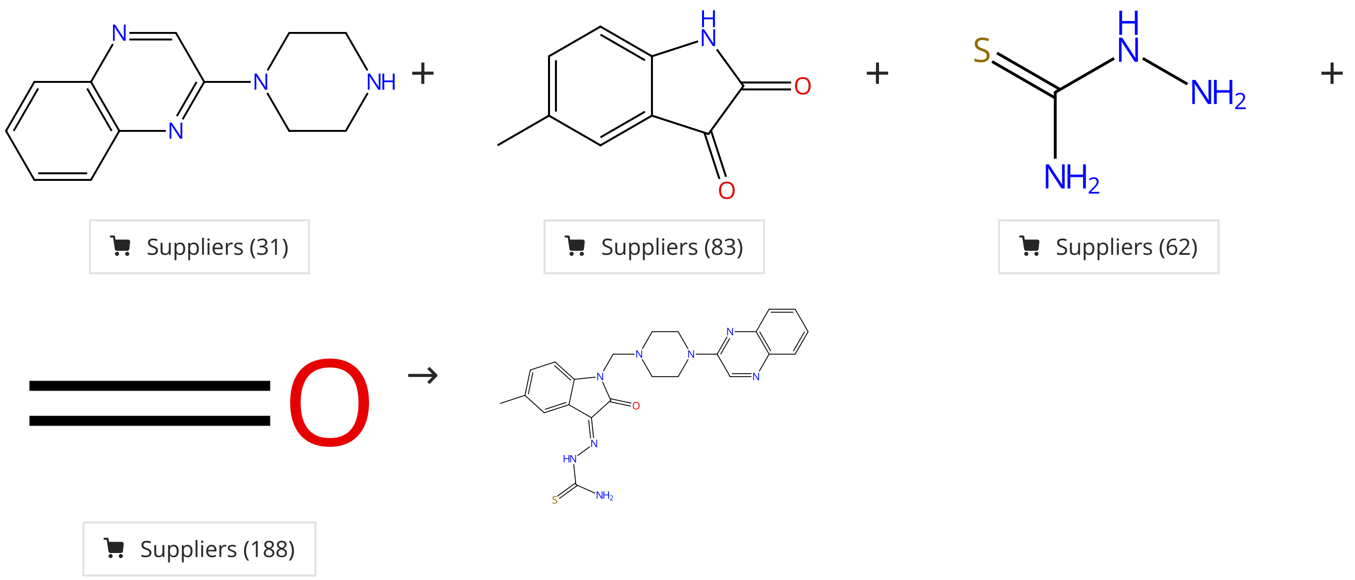
Green Synthesis and Unexpected Transformation of New Oxothiazolo[3,2-c]Pyrimidine Carbonitriles

By: Ozbil, Cagri; et al

Journal of Heterocyclic Chemistry (2025), 62(7), 458-467.

Scheme 31 (2 Reactions)

Steps: **1** Yield: **97%**



31-614-CAS-37559195	Steps: **1** Yield: **97%**
1.1	**Catalysts:** Benzenesulfonamide, 4-amino-*N*-[imino[[3-(trimethoxysilyl)propyl]amino]methyl]- (sulfonated and aluminum iron oxide supported) **Solvents:** Ethanol; 3 h, reflux

Green synthesis, anti-proliferative evaluation, docking, and MD simulations studies of novel 2-piperazinyl quinoxaline derivatives using hercynite sulfaguanidine-SA as a highly efficient and reusable nanocatalyst

By: Esam, Zohreh; et al

RSC Advances (2023), 13(36), 25229-25245.

31-310-CAS-22967840	Steps: **1** Yield: **97%**
1.1	**Catalysts:** 4-[[[[(Dimethylamino)(sulfoimino)methyl]sulfoamino](sulfoimino)methyl]imino]methyl]benzoic acid (Fe3O4 support used) **Solvents:** Ethanolamine; 2 h, reflux

Experimental Protocols

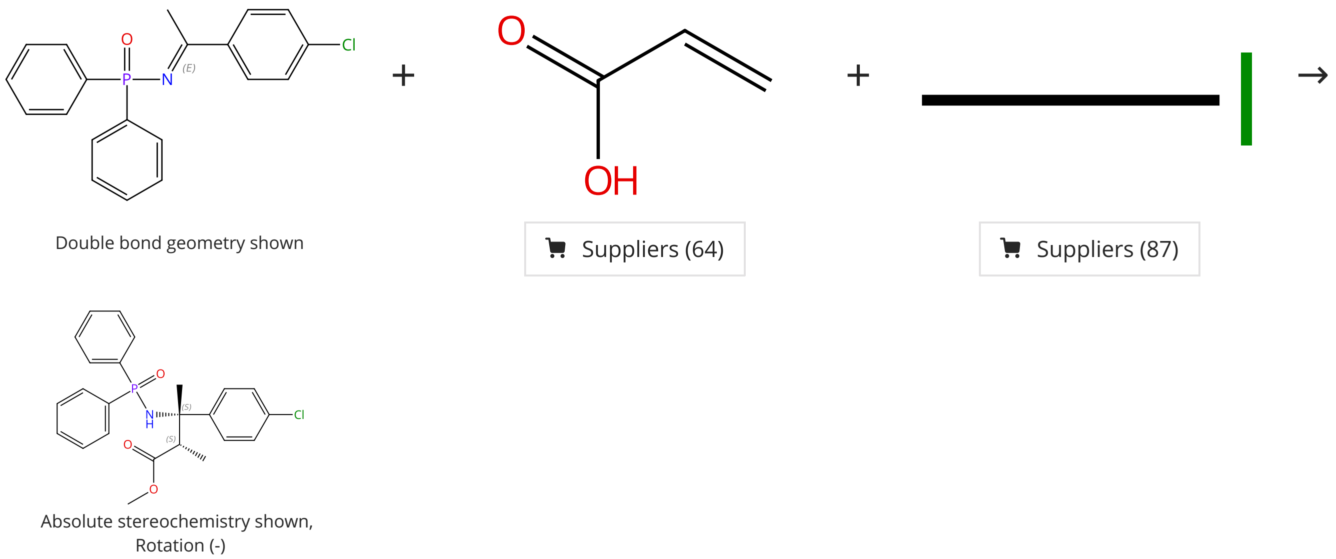
One-pot multicomponent synthesis of novel 2-(piperazin-1-yl) quinoxaline and benzimidazole derivatives, using a novel sulfamic acid functionalized Fe_3O_4 MNPs as highly effective nanocatalyst

By: Esam, Zohreh; et al

Applied Organometallic Chemistry (2021), 35(1), e6005.

Scheme 32 (1 Reaction)

Steps: **1** Yield: **97%**



31-614-CAS-38316189 Steps: **1** Yield: **97%**

1.1 **Reagents:** Trimethoxysilane
Catalysts: Cupric acetate, (+)-BINAP
Solvents: Tetrahydrofuran; 30 min, 0 °C

1.2 **Reagents:** *tert*-Butanol; 18 h, -20 °C

1.3 **Reagents:** Potassium fluoride
Solvents: Water; 10 min, rt

1.4 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 3 h, rt

1.5 **Reagents:** Water; rt

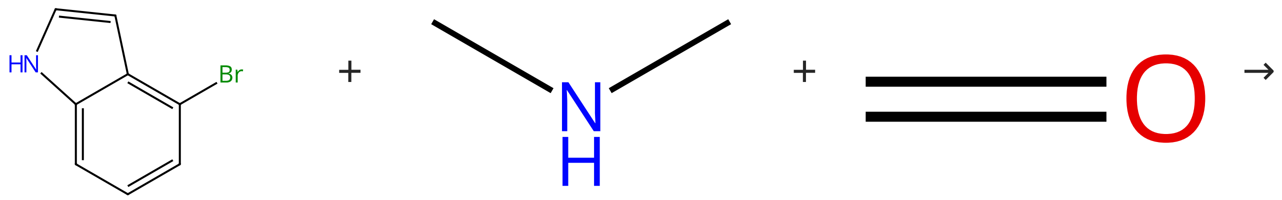
Experimental Protocols

Diastereo- and Enantioselective Reductive Mannich-type Reaction of α,β-Unsaturated Carboxylic Acids to Ketimines: A Direct Entry to Unprotected $\beta^{2,3,3}$-Amino Acids

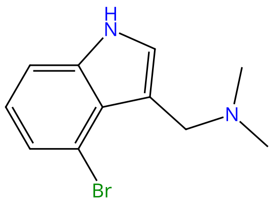
By: Suzuki, Hirotsugu; et al

Chemistry - A European Journal (2023), 29(4), e202202575.

Scheme 33 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (92) Suppliers (118) Suppliers (188)



Suppliers (31)

31-096-CAS-24404886 Steps: **1** Yield: **97%**

1.1 **Reagents:** Acetic acid
Solvents: Acetonitrile, Water; 2 h, rt

1.2 **Reagents:** Sodium hydroxide
Solvents: Water; basified, rt

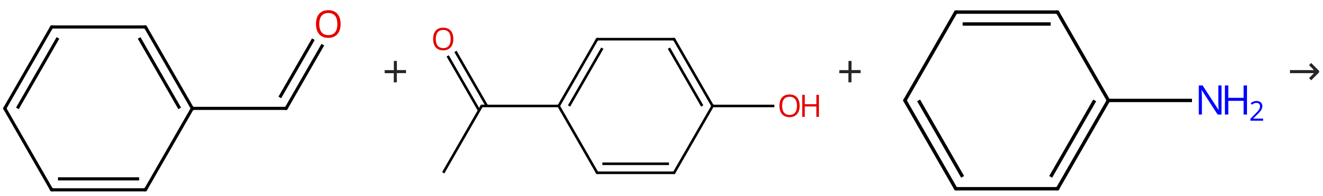
Experimental Protocols

The structural simplification of lysergic acid as a natural lead for synthesizing novel anti-Alzheimer agents

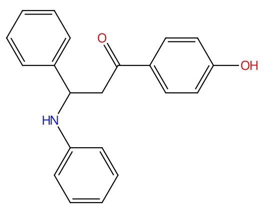
By: Alzweiri, Muhammed; et al

Bioorganic & Medicinal Chemistry Letters (2021), 47, 128205.

Scheme 34 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (69) Suppliers (122) Suppliers (113)



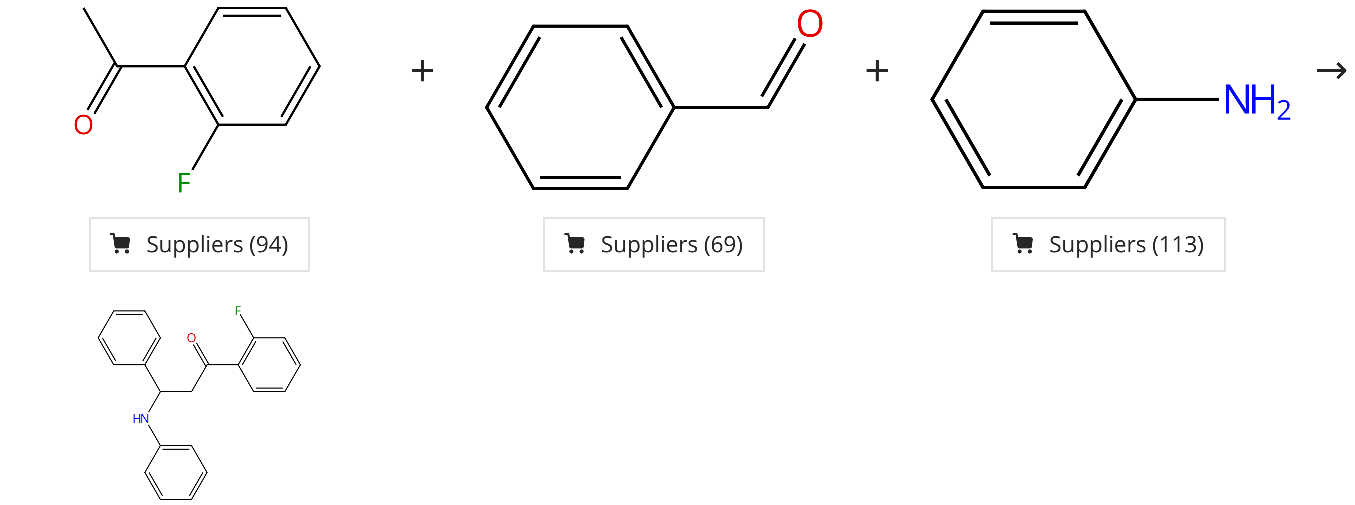
31-315-CAS-21619372	Steps: **1** Yield: **97%**
1.1	**Catalysts:** 1*H*-Imidazolium, 1-methyl-3-[2-[[(2*S*)-2-pyrrolidin ylcarbonyl]oxy]ethyl]-, 2,2,2-trifluoroacetate (1:1); 2 h, rt
Experimental Protocols	

Ionic liquid-immobilized proline(s) organocatalyst-catalyzed one-pot multi-component Mannich reaction under solvent-free condition

By: Prabhakara, M. D.; et al

Research on Chemical Intermediates (2020), 46(4), 2381-2401.

Scheme 35 (1 Reaction) Steps: **1** Yield: **97%**



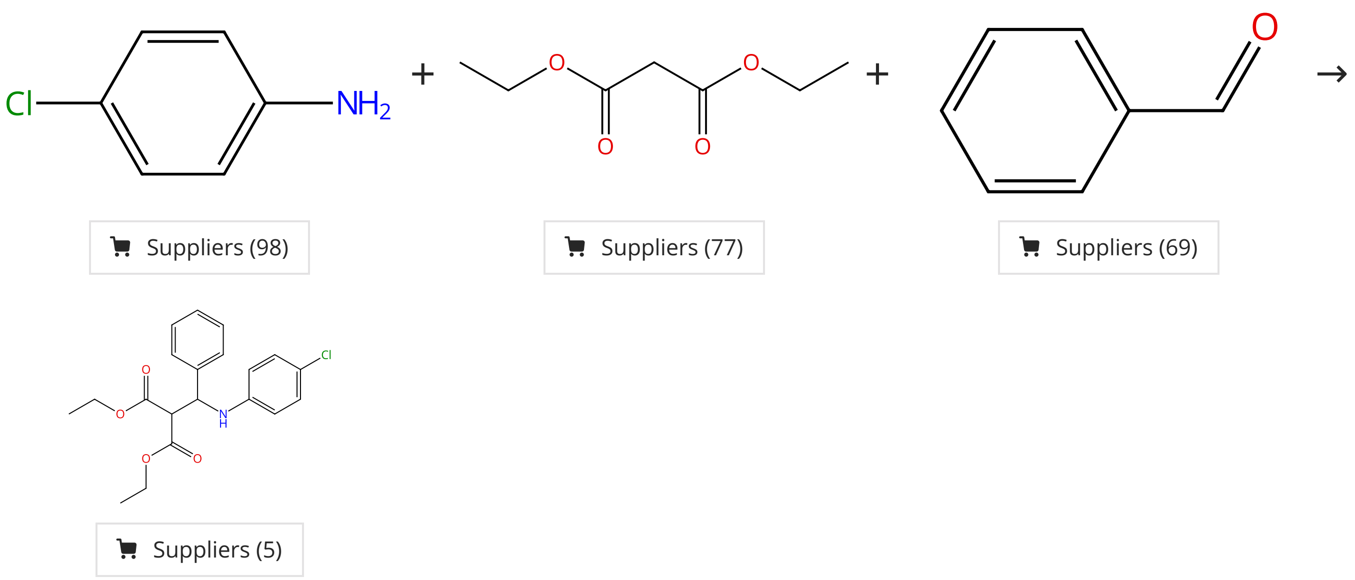
31-315-CAS-23444816	Steps: **1** Yield: **97%**
1.1	**Catalysts:** 1*H*-Imidazolium, 1-methyl-3-(3-sulfopropyl)-, (1*S*,4*R*)-7,7-dimethyl-2-oxobicyclo[2.2.1]heptane-1-methanesulfonic acid (1:1); 15 min, 25 °C
Experimental Protocols	

Mechanistic studies on counter-ionic effects of camphorsu lfonate-based ionic liquids on kinetics, thermod ynamics and stereoselectivity of β-amino carbonyl compounds

By: Sardar, Sabahat; et al

Journal of Molecular Liquids (2020), 320(Part_A), 114370.

Scheme 36 (1 Reaction) Steps: **1** Yield: **97%**



31-315-CAS-23757593	Steps: **1** Yield: **97%**
1.1	**Catalysts:** Ammonium chloride **Solvents:** Ethanol; 10 h, rt
Experimental Protocols	

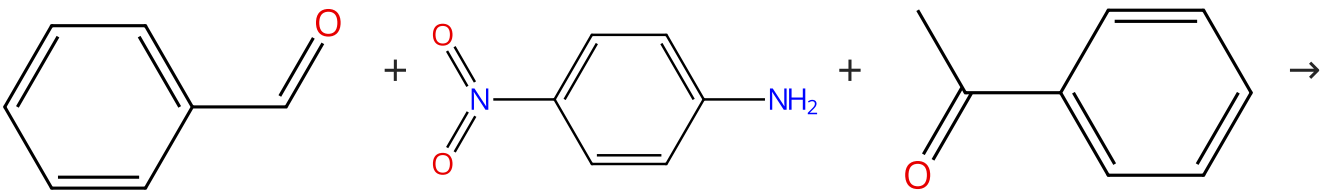
NH_4Cl Catalyzed synthesis of β-amino Esters

By: Waghmare, Smita R.

Results in Chemistry (2020), 2, 100036.

Scheme 37 (2 Reactions)

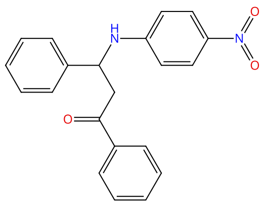
Steps: **1** Yield: **96-97%**



Suppliers (69)

Suppliers (93)

Suppliers (105)



Suppliers (4)

31-315-CAS-21619361

Steps: **1** Yield: **97%**

1.1 **Catalysts:** 1*H*-Imidazolium, 1-methyl-3-[2-[[(2*S*)-2-pyrrolidinylcarbonyl]oxy]ethyl]-, 2,2,2-trifluoroacetate (1:1); 2 h, rt

Experimental Protocols

Ionic liquid-immobilized proline(s) organocatalyst-catalyzed one-pot multi-component Mannich reaction under solvent-free condition

By: Prabhakara, M. D.; et al

Research on Chemical Intermediates (2020), 46(4), 2381-2401.

31-614-CAS-24768620

Steps: **1** Yield: **96%**

1.1 **Catalysts:** 2747176-70-9; 15 - 30 min, rt

Experimental Protocols

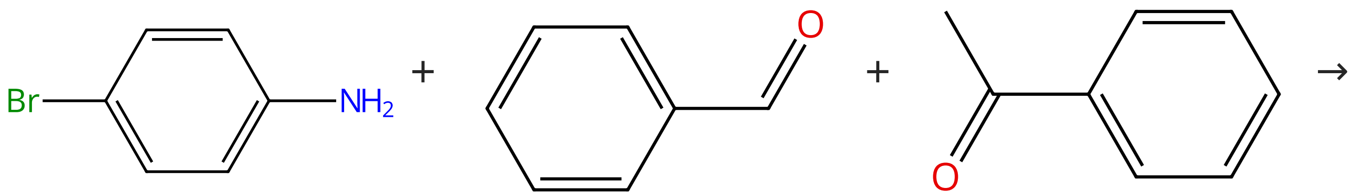
Bifunctional C_2-symmetric ionic liquid-supported (S)-proline as a recyclable organocatalyst for Mannich reactions in neat condition

By: Davanagere, Prabhakara Madivalappa; et al

Results in Chemistry (2021), 3, 100152.

Scheme 38 (1 Reaction)

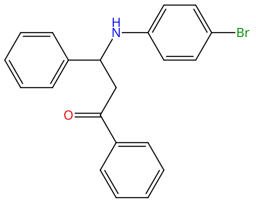
Steps: **1** Yield: **97%**



Suppliers (83)

Suppliers (69)

Suppliers (105)



Suppliers (4)

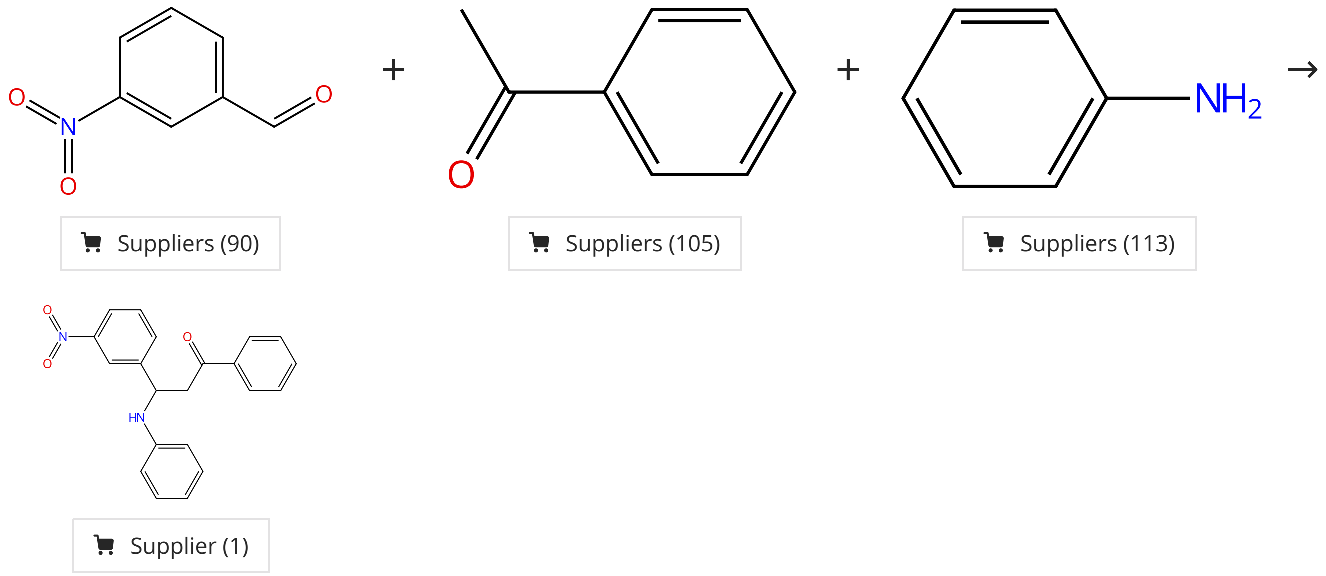
31-315-CAS-22634029	Steps: **1** Yield: **97%**
1.1	**Catalysts:** Cadmium (reaction product with saponin, support on Fe3O4) **Solvents:** Ethanol; 3 h, rt
Experimental Protocols	

Fe_3O_4@saponin/Cd: a novel magnetic nano-catalyst for the synthesis of β-aminoketone derivatives

By: Kiani, Marzieh; et al

Journal of the Iranian Chemical Society (2020), 17(9), 2243-2256.

Scheme 39 (1 Reaction) Steps: **1** Yield: **97%**



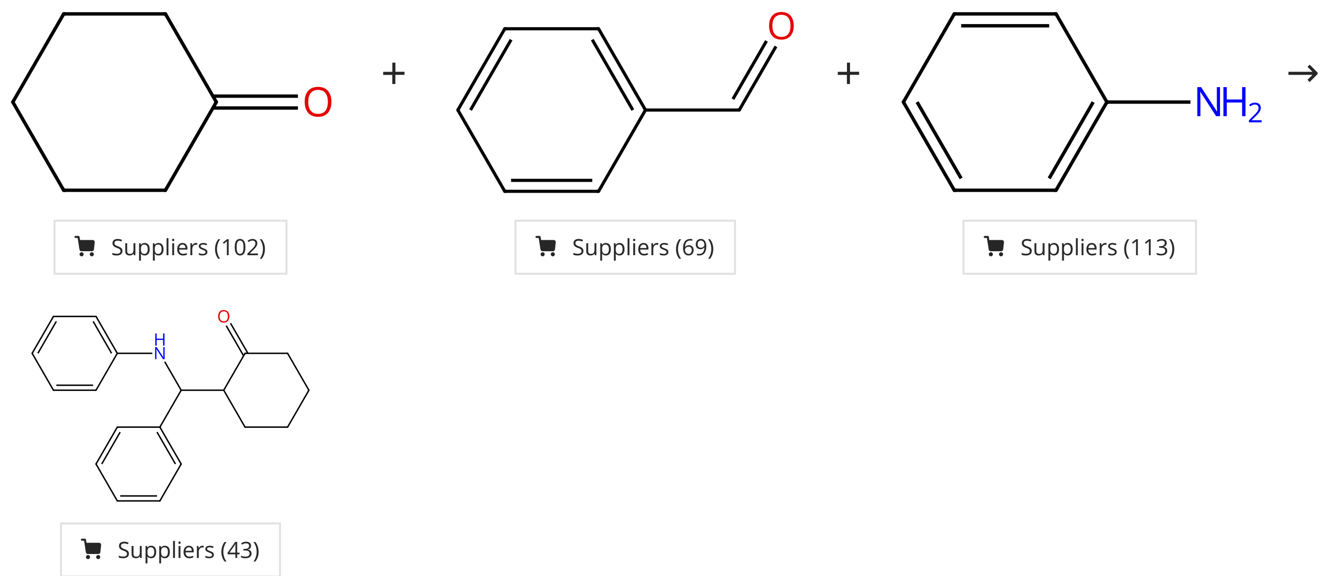
31-614-CAS-31159494	Steps: **1** Yield: **97%**
1.1	**Catalysts:** Titanium (supported on MCM-41) **Solvents:** Ethanol; 5 h, rt
Experimental Protocols	

An Efficient Synthesis of Heterogeneous and Hard Bound Ti^{IV}-MCM-41 Catalyzed Mannich Bases and π-Conjugated Imines

By: Muthumanickam, Shenbagapushpam; et al

ChemistrySelect (2021), 6(48), 14071-14076.

Scheme 40 (1 Reaction) Steps: **1** Yield: **97%**



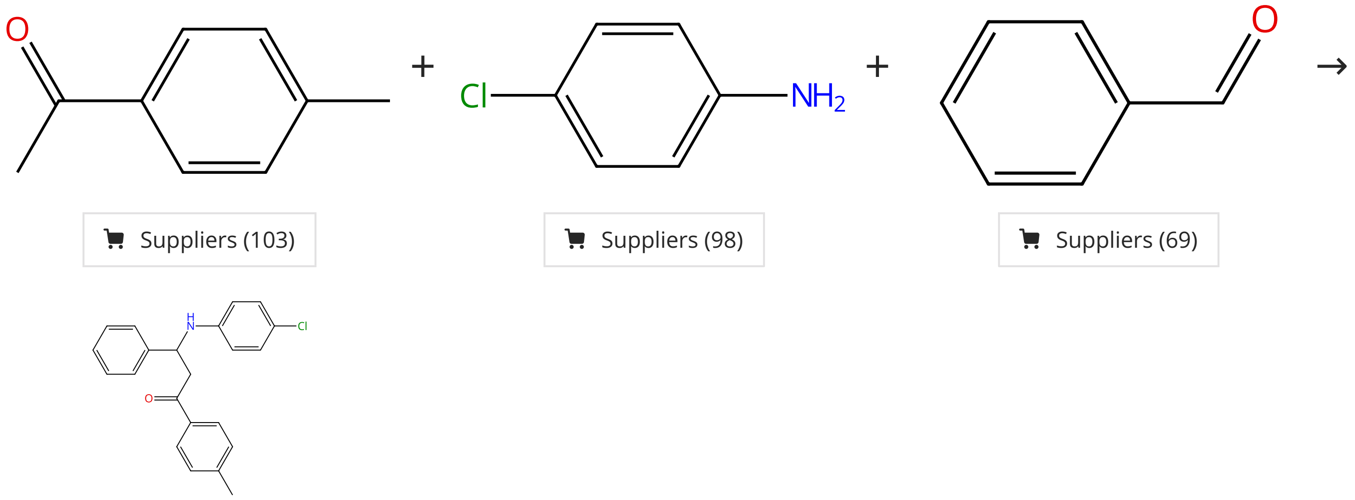
31-614-CAS-41264716	Steps: **1** Yield: **97%**
1.1	**Catalysts:** Oxirane, (chloromethyl)-, homopolymer, ether with D-mannitol (6:1) (methylated, azidized and hydrogenated) **Solvents:** Water; 3 min, 30 °C
Experimental Protocols	

Design of Dendritic Foldamers as Catalysts for Organic Synthesis

By: Baby, Sherlymole P.; et al

Current Organocatalysis (2024), 11(3), 214-231.

Scheme 41 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (103) Suppliers (98) Suppliers (69)

Suppliers (4)

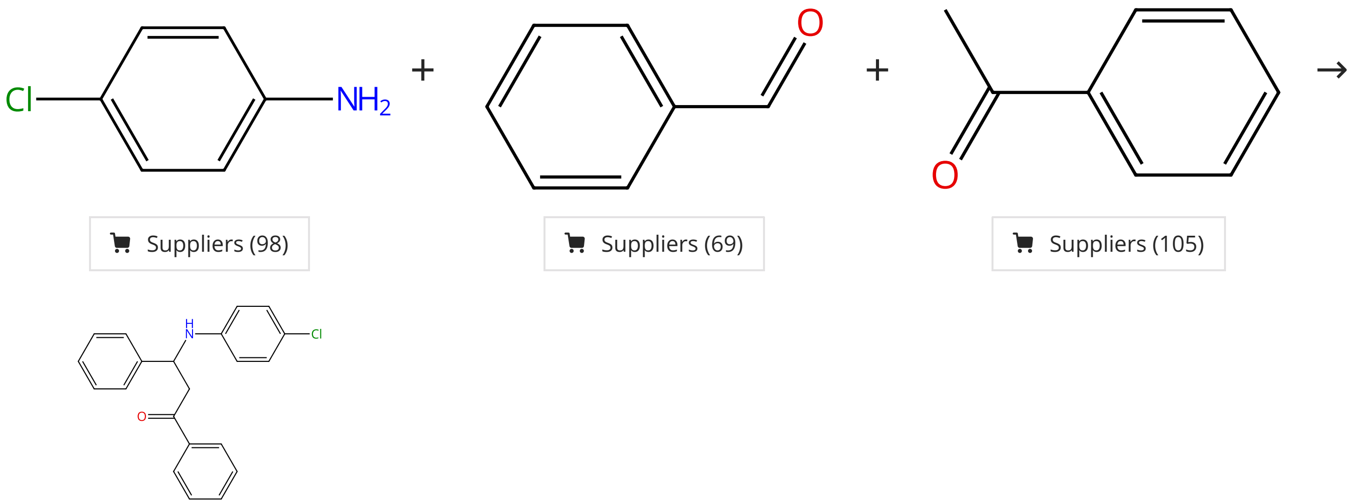
31-614-CAS-24768608	Steps: **1** Yield: **97%**
1.1	**Catalysts:** 2747176-70-9; 33 min, rt
Experimental Protocols	

Bifunctional C_2-symmetric ionic liquid-supported (S)-proline as a recyclable organocatalyst for Mannich reactions in neat condition

By: Davanagere, Prabhakara Madivalappa; et al

Results in Chemistry (2021), 3, 100152.

Scheme 42 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (98) Suppliers (69) Suppliers (105)

Suppliers (3)

31-315-CAS-21619356	Steps: **1** Yield: **97%**
1.1	**Catalysts:** 1*H*-Imidazolium, 1-methyl-3-[2-[[(2*S*)-2-pyrrolidinylcarbonyl]oxy]ethyl]-, 2,2,2-trifluoroacetate (1:1); 2 h, rt
Experimental Protocols	

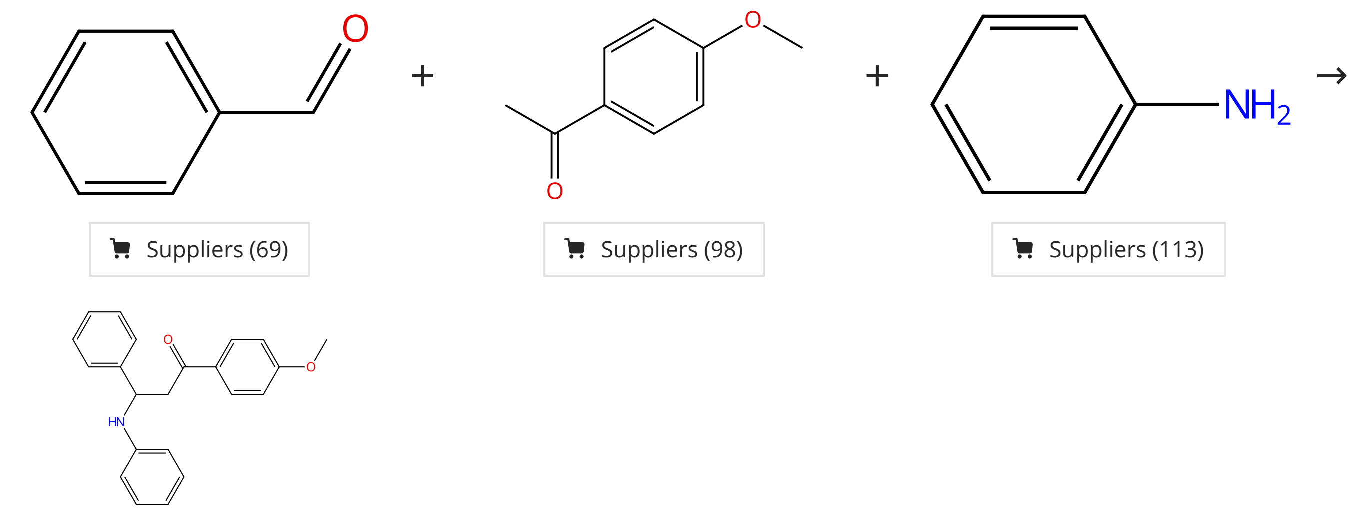
Ionic liquid-immobilized proline(s) organocatalyst-catalyzed one-pot multi-component Mannich reaction under solvent-free condition

By: Prabhakara, M. D.; et al

Research on Chemical Intermediates (2020), 46(4), 2381-2401.

Scheme 43 (1 Reaction)

Steps: **1** Yield: **96%**



Suppliers (69) Suppliers (98) Suppliers (113)

Suppliers (4)

31-315-CAS-21619375	Steps: **1** Yield: **96%**
1.1	**Catalysts:** 1*H*-Imidazolium, 1-methyl-3-[2-[[(2*S*)-2-pyrrolidinylcarbonyl]oxy]ethyl]-, 2,2,2-trifluoroacetate (1:1); 2 h, rt
Experimental Protocols	

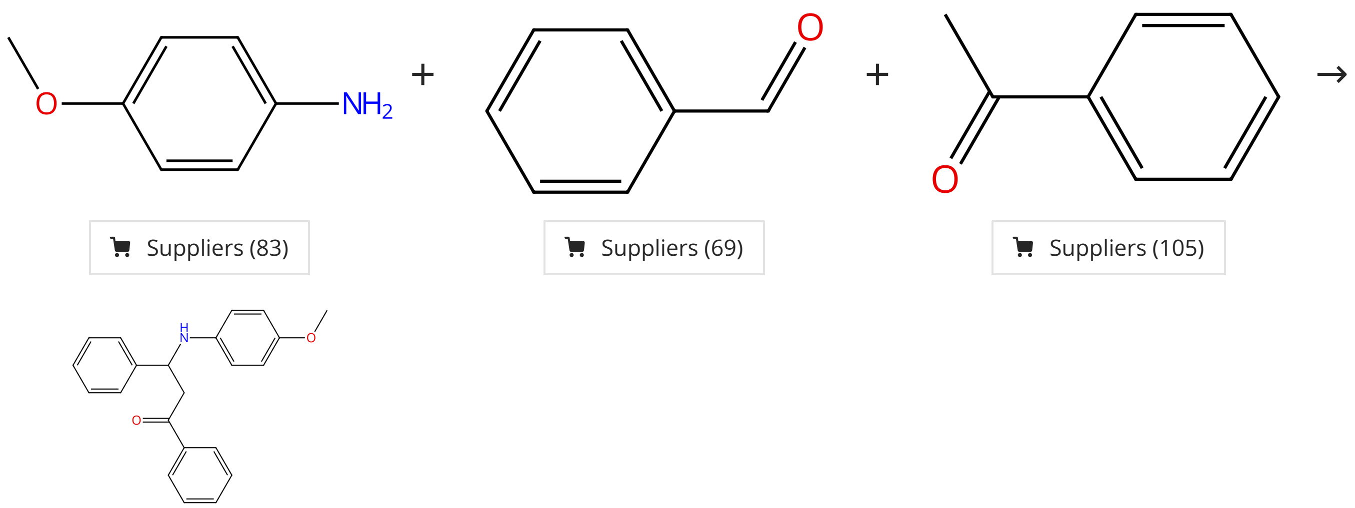
Ionic liquid-immobilized proline(s) organocatalyst-catalyzed one-pot multi-component Mannich reaction under solvent-free condition

By: Prabhakara, M. D.; et al

Research on Chemical Intermediates (2020), 46(4), 2381-2401.

Scheme 44 (1 Reaction)

Steps: **1** Yield: **96%**



Suppliers (83) Suppliers (69) Suppliers (105)

Suppliers (10)

31-315-CAS-21619362 Steps: **1** Yield: **96%**	**Ionic liquid-immobilized proline(s) organocatalyst-catalyzed one-pot multi-component Mannich reaction under solvent-free condition**
1.1 **Catalysts:** 1*H*-Imidazolium, 1-methyl-3-[2-[[(2*S*)-2-pyrrolidinylcarbonyl]oxy]ethyl]-, 2,2,2-trifluoroacetate (1:1); 2 h, rt	By: Prabhakara, M. D.; et al
Experimental Protocols	Research on Chemical Intermediates (2020), 46(4), 2381-2401.